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1 **Medical Image Segmentation in a Multiple**
2 **Labelers Context: Application to the Study**
3 **of Histopathology**

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Medical Image Segmentation in a Multiple Labelers Context: Application to the Study of Histopathology

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Segmentación de Imágenes Médicas en un Contexto de Múltiples Anotadores: Aplicación al Estudio de Histopatologías

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ABSTRACT

60 Breast cancer remains one of the deadliest diseases affecting women worldwide,
61 making accurate diagnosis and treatment planning crucial for improving patient
62 outcomes. Semantic segmentation of histopathology images plays a vital role in
63 this process, as it enables precise identification and analysis of tissue structures.
64 However, the current approaches face significant challenges in real-world
65 scenarios where multiple annotators with varying expertise and reliability
66 contribute to the labeling process. This crowdsourcing paradigm, while
67 cost-effective, introduces inconsistencies that can compromise the quality of the
68 segmentation results.

69 This thesis addresses two fundamental challenges in medical image segmentation:
70 the inherent uncertainty in multiple annotator scenarios and the need for robust
71 segmentation models that can handle varying image quality and annotator
72 expertise. These challenges arise from the complex nature of histopathology
73 images, where tissue structures can be ambiguous, and the varying levels of
74 expertise among annotators lead to inconsistent labeling patterns. Additionally,
75 the high cost and time requirements for expert annotations create a need for
76 efficient solutions that can work with crowdsourced data.

77 To address these challenges, this thesis proposes a comprehensive approach that
78 combines novel loss functions with robust deep learning architectures to handle
79 multiple annotator scenarios effectively.

80 The first objective introduces a novel loss function that implicitly infers optimal
81 segmentation while assessing labeler reliability without requiring explicit
82 performance inputs. This approach differs from conventional methods by
83 eliminating the need for direct supervision of labeler performance, instead using
84 an intermediate reliability map that enables the model to prioritize information
85 from reliable labelers while disregarding noisy labels. The mathematical
86 foundation of this approach lies in the probabilistic modeling of annotator
87 reliability across the input space.

88 The second objective focuses on the development of a robust segmentation
89 architecture that combines the strengths of U-shaped deep learning models with
90 the proposed reliability-aware loss function. This combination enables the model
91 to learn both the segmentation task and the reliability patterns simultaneously,
92 leading to more accurate and consistent results. The architecture's key innovation
93 lies in its ability to adapt to varying image quality and annotator expertise levels
94 without requiring explicit performance metrics.

95 The third objective involves validating the proposed approach through extensive
96 experimentation on both synthetic datasets and real-world histopathology images.
97 The evaluation demonstrates superior performance compared to state-of-the-art
98 approaches in terms of segmentation accuracy and uncertainty quantification. The
99 results show that the proposed method can effectively handle inconsistent
100 annotations while maintaining high segmentation accuracy, making it particularly
101 valuable for clinical applications.

102 The research makes significant contributions to the field by providing a robust
103 solution for handling inconsistent annotations in medical image segmentation,
104 particularly valuable in histopathology where expert annotations are expensive
105 and time-consuming to obtain. The proposed approach shows promising results in
106 reducing annotation costs while maintaining high segmentation accuracy, making
107 it particularly relevant for clinical applications where precise segmentation of
108 tissue structures is crucial for diagnosis and treatment planning. This work opens

109 new research directions in the areas of crowdsourced medical image analysis,
110 uncertainty quantification in deep learning, and the development of more efficient
111 annotation protocols for medical imaging.

112 **Keywords:** Crowdsourcing, Multiple Annotators, Histopathology, Breast Cancer,
113 Semantic Image Segmentation, Gaussian Processes, Deep Learning, Medical Image
114 Analysis

116 El cáncer de mama sigue siendo una de las enfermedades más mortales en la
117 población femenina en todo el mundo, lo que hace que el diagnóstico preciso y la
118 planificación del tratamiento sean cruciales para mejorar las condiciones de los
119 pacientes. La segmentación semántica de imágenes histopatológicas juega un
120 papel vital en este proceso, ya que permite la identificación y análisis preciso de
121 estructuras tisulares. Sin embargo, los enfoques actuales enfrentan desafíos
122 significativos en escenarios del mundo real donde múltiples anotadores con
123 diferentes niveles de experiencia y confiabilidad contribuyen al proceso de
124 etiquetado. Este paradigma de crowdsourcing, aunque rentable, introduce
125 inconsistencias que pueden comprometer la calidad de los resultados de
126 segmentación.

127 Esta tesis aborda dos desafíos fundamentales en la segmentación de imágenes
128 médicas: la incertidumbre inherente en escenarios de múltiples anotadores y la
129 necesidad de modelos de segmentación robustos que puedan manejar la calidad
130 variable de las imágenes y la experiencia de los anotadores. Estos desafíos surgen
131 de la naturaleza compleja de las imágenes histopatológicas, donde las estructuras
132 tisulares pueden ser ambiguas, y los diferentes niveles de experiencia entre los
133 anotadores conducen a patrones de etiquetado inconsistentes. Además, el alto
134 costo y los requisitos de tiempo para las anotaciones de expertos crean la
135 necesidad de soluciones eficientes que puedan trabajar con datos crowdsourced.

El primer objetivo introduce una función de pérdida novedosa que infiere implícitamente la segmentación óptima mientras evalúa la confiabilidad de los etiquetadores sin requerir entradas explícitas sobre su rendimiento. Este enfoque difiere de los métodos convencionales al eliminar la necesidad de supervisión directa del rendimiento de los etiquetadores, utilizando en su lugar un mapa de confiabilidad intermedio que permite al modelo priorizar la información de etiquetadores confiables mientras descarta etiquetas ruidosas. La base matemática de este enfoque radica en el modelado probabilístico de la confiabilidad de los anotadores a través del espacio de entrada.

El segundo objetivo se centra en el desarrollo de una arquitectura de segmentación robusta que combina las fortalezas de los modelos de aprendizaje profundo en forma de U con la función de pérdida propuesta que tiene en cuenta la confiabilidad. Esta combinación permite que el modelo aprenda tanto la tarea de segmentación como los patrones de confiabilidad simultáneamente, lo que lleva a resultados más precisos y consistentes. La innovación clave de la arquitectura radica en su capacidad para adaptarse a diferentes niveles de calidad de imagen y experiencia de los anotadores sin requerir métricas de rendimiento explícitas.

El tercer objetivo valida el enfoque propuesto a través de experimentación exhaustiva tanto en conjuntos de datos sintéticos como en imágenes histopatológicas del mundo real. La evaluación demuestra un rendimiento superior en comparación con los enfoques más avanzados en términos de precisión de segmentación y cuantificación de incertidumbre. Los resultados muestran que el método propuesto puede manejar efectivamente anotaciones inconsistentes mientras mantiene una alta precisión de segmentación, lo que lo hace particularmente valioso para aplicaciones clínicas.

La investigación hace contribuciones significativas al campo al proporcionar una solución robusta para manejar anotaciones inconsistentes en la segmentación de imágenes médicas, particularmente valiosa en histopatología donde las anotaciones de expertos son costosas y requieren mucho tiempo para obtenerlas.

165 El enfoque propuesto muestra resultados prometedores en la reducción de costos
166 de anotación mientras mantiene una alta precisión de segmentación, lo que lo hace
167 particularmente relevante para aplicaciones clínicas donde la segmentación precisa
168 de estructuras tisulares es crucial para el diagnóstico y la planificación del
169 tratamiento. Este trabajo abre nuevas direcciones de investigación en las áreas de
170 análisis de imágenes médicas crowdsourced, cuantificación de incertidumbre en
171 aprendizaje profundo y el desarrollo de protocolos de anotación más eficientes
172 para imágenes médicas.

173 **Palabras clave:** Crowdsourcing, Múltiples Anotadores, Histopatología, Cáncer de
174 Mama, Segmentación Semántica de Imágenes, Procesos Gaussianos, Aprendizaje
175 Profundo, Análisis de Imágenes Médicas

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ABBREVIATIONS

- 182 **CAD** Computer-Aided Diagnosis 2, 5, 8
- 183 **CCGP** Correlated Chained Gaussian Processes 18
- 184 **CCGPMA** Correlated Chained Gaussian Processes for Multiple Annotators xxii, 18, 19,
185 53, 55–57, 59, 95
- 186 **CE** Cross Entropy 62, 65
- 187 **CGP** Chained Gaussian Processes 18, 45, 46
- 188 **CKA** Centered Kernel Alignment 44
- 189 **CNN** Convolutional Neural Networks 3, 15, 67, 95
- 190 **CR** Crowdsourcing 16, 17
- 191 **CT** Computed Tomography 1, 2, 12
- 192 **ELBO** Evidence Lower Bound 19
- 193 **EM** Expectation-Maximization 15, 20
- 194 **GCE** Generalized Cross Entropy 62, 65, 71, 84
- 195 **GCECDL** Generalized Cross-Entropy-based Chained Deep Learning 21
- 196 **GP** Gaussian Process 46, 58
- 197 **GPMC** Gaussian Process Monte Carlo 48
- 198 **GPR** Gaussian Process Regression 48

- 199 **ISS** Image Semantic Segmentation 2, 5, 8, 10, 14, 21, 23, 37, 93, 95
- 200 **KAAR** Kernel Alignment-Based Annotator Relevance Analysis 43, 44, 46, 57
- 201 **LF** Latent Function 18
- 202 **LKAAR** Localized Kernel Alignment-Based Annotator Relevance Analysis 44–46, 57
- 203 **MAE** Mean Absolute Error 62, 65, 71
- 204 **MITs** Medical Imaging Techniques 1
- 205 **ML** Machine Learning 10, 12, 21
- 206 **MRI** Magnetic Resonance Imaging 1, 2
- 207 **MV** Majority Voting 10, 12, 20, 22
- 208 **OCR** Optical Character Recognition 12
- 209 **PET** Positron Emission Tomography 14
- 210 **RCDNN** Regularized Chained Deep Neural Network 47, 57
- 211 **ROI** Region of Interest 2, 8, 16, 31, 87
- 212 **SGPMC** Sparse Gaussian Process Monte Carlo 48
- 213 **SGPR** Sparse Gaussian Process Regression 48
- 214 **SLFM** Semi-Parametric Latent Factor Model 18, 51, 52
- 215 **SS** Semantic segmentation 3, 67
- 216 **STAPLE** Simultaneous Truth and Performance Level Estimation 13–15, 38
- 217 **TNBC** Triple-Negative Breast Cancer xxiii, xxiv, 41, 84, 87–91
- 218 **WSI** Whole Slide Imaging xxii, 1, 5, 8, 16, 29, 31, 40, 87

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221

222

CHAPTER

ONE

223

INTRODUCTION

224

1.1 Motivation

225 Since Roentgen's discovery of X-rays in 1895, medical imaging has advanced
226 significantly, with modalities like radionuclide imaging, ultrasound, **Computed**
227 **Tomography (CT)**, **Magnetic Resonance Imaging (MRI)**, and digital radiography
228 emerging over the past 50 years. Modern imaging extends beyond image
229 production to include processing, display, storage, transmission and analysis. [?].
230 Other **Medical Imaging Techniques (MITs)** have arose during the last decades,
231 some of them implying only the examination of certain pieces or tissues instead of
232 complete patients, like histopathological images, which are images of tissue
233 samples obtained from biopsies or surgical resections and are widely used for the
234 diagnosis of diseases like cancer through **Whole Slide Imaging (WSI)** scanners [?].

235 Along with the advances in technologies for medical images acquisition,
236 computational technologies on pattern recognition and artificial intelligence have

also emerged, allowing the development of **Computer-Aided Diagnosis (CAD)** systems based on Machine Learning algorithms. These systems aim to assist physicians in the diagnosis and treatment of diseases, by providing a second opinion or by automating the analysis of medical images. [?]. One of the most common tasks in which machine learning technologies are being used in the universe of medical images is **Image Semantic Segmentation (ISS)**, which consists of assigning a label to each pixel in an image according to the object it belongs to. This is crucial for the development of **CAD** systems, as it allows the identification of **Region of Interest (ROI)** in the images, which can be used to detect and classify diseases [?].

The application of Machine Learning in medical imaging has grown significantly, including classification, segmentation, anomaly detection, super-resolution, image registration, and synthetic image generation [?]. Among imaging modalities, X-rays and **CT** scans are widely used for classification and anomaly detection, especially in pulmonary and oncological applications. **MRI** and ultrasound play a crucial role in segmentation and resolution enhancement, while PET/SPECT imaging is essential for anomaly detection in oncology and neurodegenerative diseases [?]. Histopathology is rapidly gaining prominence, particularly in segmentation and feature extraction, where AI-driven techniques aid in automated cancer diagnosis and tissue structure analysis. The integration of Deep Learning in histological image processing is revolutionizing pathology, enabling more precise and efficient diagnostics. A brief comparison of the tasks and medical image types based on counted referenced terms occurrences from recent literature review, can be seen in Figure ?? [?], [?], [?], [?], [?]

Throughout the time that medical images and computational techniques have been developed, a coevolution has occurred, where the complexity of the tasks and the capabilities of the computational techniques have been increasing [?]. Initially, these needs were covered with simple morphological transformations, which implied no training process or elaborated optimization, this is an advantage in terms of development costs but has limited performance. However, as the

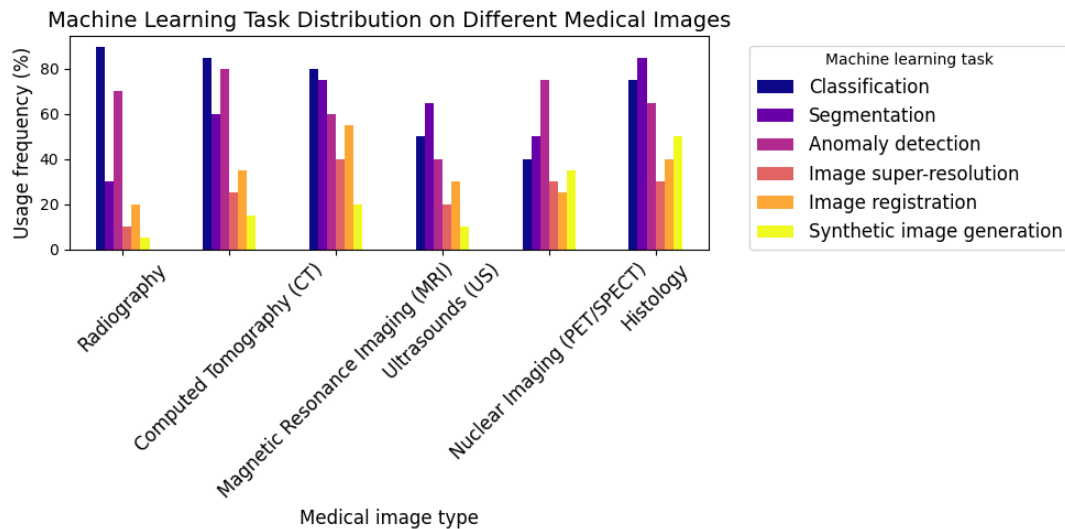


Figure 1-1 Estimation of the popularity of tasks and medical image types based on recent literature review (count of referenced terms.)

complexity of the tasks increased, the need for more sophisticated techniques arose, leading to the application of advanced statistical tools and Machine Learning algorithms like Support Vector Machines, Decision Trees, and SGD Neural Networks [?]. The coevolution of advances in medical image acquisition, computational power (i.e. Moore's law) and statistical/mathematical techniques have led to a convergence for merging state of the art algorithms with medical imaging [?]. Figure ?? shows a timeline of coevolution between some conspicuous advances in computational pattern recognition and its medical applications in different scopes (besides medical imaging) [?].

Convolutional Neural Networks (CNN) have been widely used in Semantic segmentation (SS), as they have outperformed traditional Machine Learning algorithms both medical and non medical images [?] [?]. However, most CNN architectures are deep, which introduces the requirement of large amounts of training data. This introduces a problem since the acquisition and annotation of medical images are expensive and time-consuming processes. This is especially true for ISS tasks, as they require pixel-level annotations, which is taxing in terms of cost, time and the logistics involved [?]. Other modalities face this problem

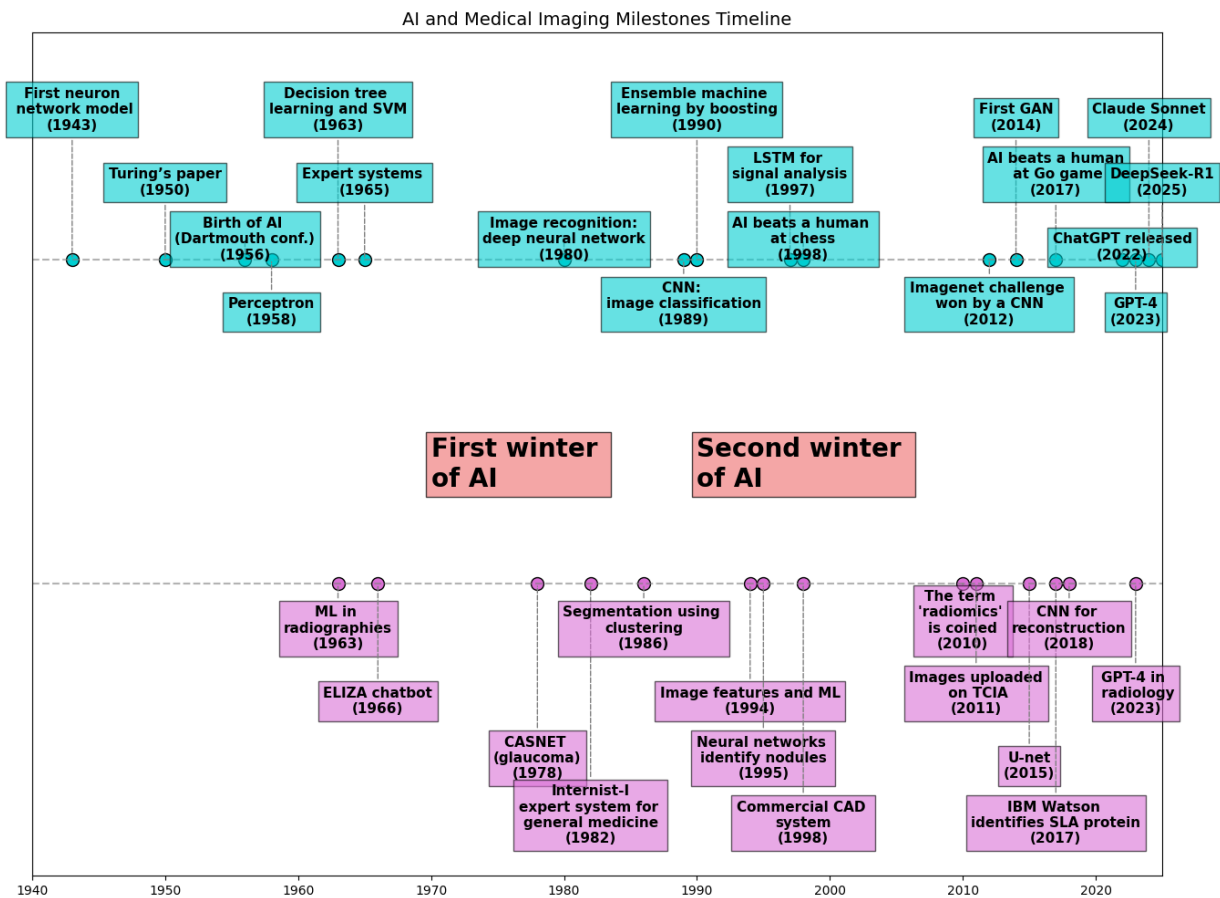


Figure 1-2 AI and Machine Learning in medical imaging recent timeline.

284 through less expensive annotation strategies like bounding boxes or anatomical
285 landmarks for being used in a semi-supervised strategy [?].

286 However, many medical images datasets contain a high variability in class sizes
287 and variations in colors, which is specially noticeable in histopathological images
288 because of the usage of different staining techniques and other factors which can
289 affect the color of the images (see section ??). This variability can lead to a
290 significant loss of efficiency of Machine Learning models when using a mixed
291 supervision strategy, as the model can be biased towards the most common
292 classes or colors in the dataset [?].

293 This is where other solutions arise to tackle the problem of the weak image
294 annotation while maintaining low costs. One of these solutions is the *crowdsourcing*
295 strategy, which consists of having multiple annotators labeling the same image,
296 and then combining the labels to obtain a consensus label [?]. This strategy can
297 lead to a labeling cost reduction when different levels of expertise are combined,
298 since the crowd may be composed of both experts and laymen, the latter being
299 less expensive to hire [?].

300 Recently, diagnosis, prognosis and treatment of cancer have heavily relied on
301 histopathology, where tissue samples are obtained through biopsies or surgical
302 resections and critical information that helps pathologists determine the presence
303 and severity of malignancies [?]. The segmentation of histopathological images
304 enables precise identification of structures such as nuclei, glands, and tumors,
305 which are essential for assessing disease progression and treatment response [?].
306 Accurate segmentation is particularly crucial in digital pathology, where
307 whole-slide images (WSI) are analyzed using AI-powered CAD systems to support
308 clinical decision-making [?].

309 A major challenge in histopathological image segmentation arises from the
310 variability in annotations provided by different pathologists. Unlike natural
311 images, where object boundaries are often well-defined, histological structures
312 may have ambiguous borders, leading to inconsistencies among annotators [?].

313 Because of this, crowdsourcing labeling is one of the most popular approaches, as
314 illustrated in Figure ??, ¹. These discrepancies highlight the need for models that
315 can handle annotation uncertainty effectively. Leveraging crowdsourcing
316 strategies and Machine Learning techniques that infer annotator reliability can
317 enhance segmentation performance while reducing costs.

¹obtained from a real world Triple Negative Breast Cancer (TNBC) dataset published in [?]

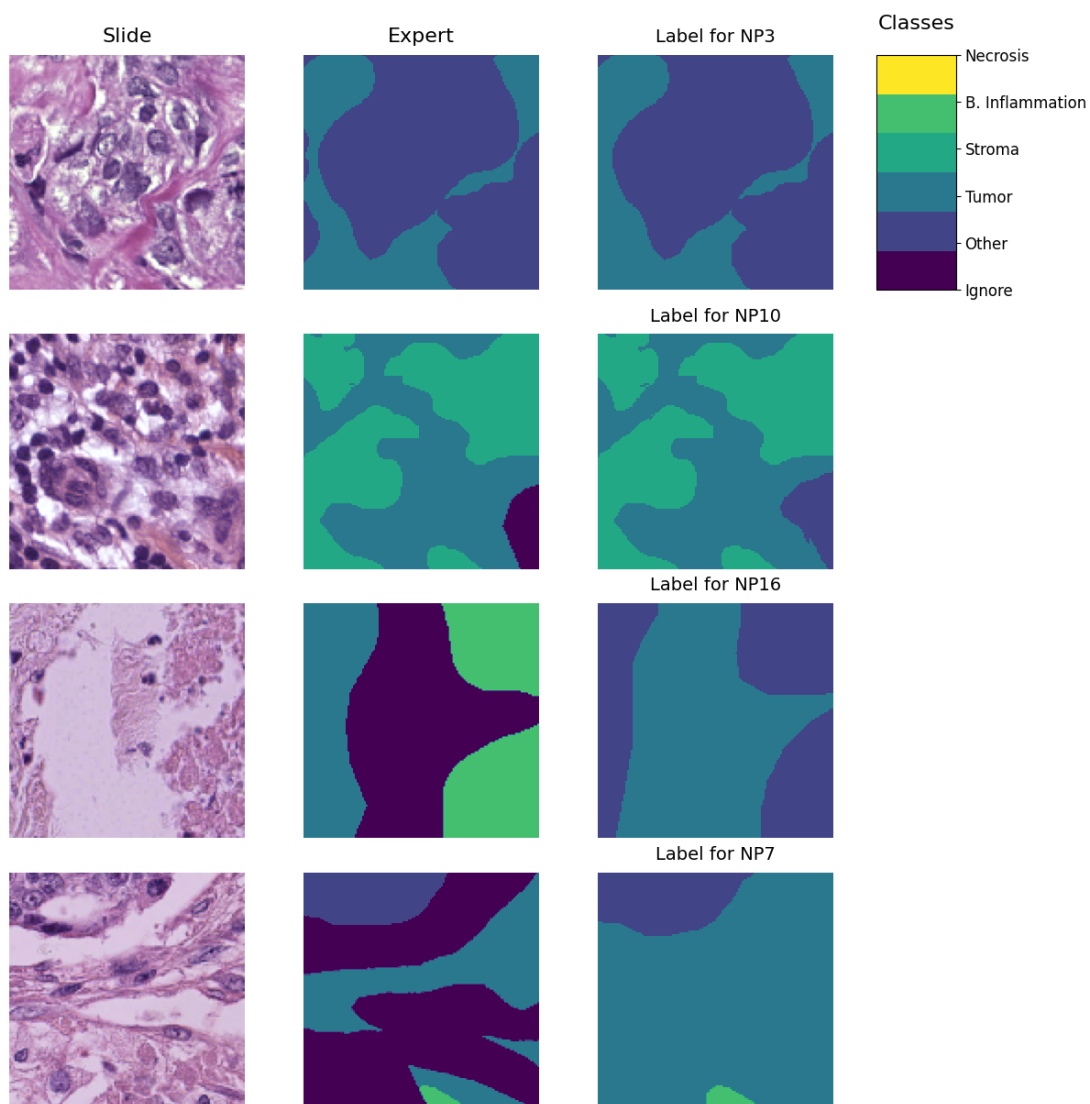


Figure 1-3 Example of a histopathological image segmented by multiple annotators, illustrating variations in label assignment.

1.2 Problem Statement

Throughout the development of medical technology and CAD, the task of ISS has become a crucial step in delivering precise diagnosis and treatment planning [?]. Particularly, in the area of histopathological studies, the usage of Whole Slide Images (WSI) is rather common since this method delivers high quality imaging and allows for the diagnosis of diseases like cancer [?].

ISS task consists of assigning a label to each pixel in an image according to the object it belongs to. Accurate segmentation is essential for the development of CAD systems, as it allows the identification of regions of interest (ROI) in the images, which can be used to detect and classify diseases and hence, treatment planning [?]. However, modern computational solutions for ISS tasks involve the use of deep learning, which mostly rely large amounts of labeled data to train the models on supervised learning techniques. This means that the model is trained on a dataset with ground-truth labels, which are assumed to be correct and consistent across all samples. In practice, this assumption is often violated due to the high technical complexity of labeling these segments ².

The process of labeling medical images is often managed with the help of specialized software tools that allow the annotators to draw the regions, delivering a standard format for the labeled masks [?]. Despite the help of these tools, the labeling process in WSI can have high costs, as it requires long hours of work from specialized personnel. In the pursuit of reducing costs, the labeling processes is often done by multiple labelers with varying levels of expertise, equalizing the cost of the labeling process. However, this strategy can lead to inconsistent labels, as the consensus between the labelers may not be exact due to the diversity in depth of knowledge and experience of the labelers [?]. These inconsistencies are discussed in the subsections ?? and ??.

²compared to a more trivial task like image classification on ordinary an well known classes like MNIST

1.2.1 Variability in Expertise Levels

One of the primary sources of inter-observer variability in medical image segmentation is the difference in expertise levels among annotators [?]. Experienced radiologists and pathologists tend to produce highly precise annotations, whereas novice labelers may introduce systematic biases due to their limited familiarity with subtle image features. Studies have demonstrated that annotation accuracy *tends* to improve with experience, yet medical institutions often rely on a mix of annotators to manage costs and workload distribution [?].

The training background of annotators and institutional guidelines play a crucial role in shaping labeling practices. Different medical schools and hospitals may adopt distinct segmentation protocols, leading to inconsistencies when datasets are combined from multiple sources [?]. For example, some institutions may emphasize conservative delineation of tumor boundaries, while others adopt a more inclusive approach. Such variations contribute to systematic biases in medical image datasets [?].

Medical images frequently contain complex structures which are difficult to distinguish, making segmentation inherently subjective. For instance, tumor margins in histopathological slides may not have well-defined edges, leading to variations in how different annotators delineate the regions of interest [?]. These discrepancies arise not only from technical expertise but also from differences in perception and interpretation.

1.2.2 Technical Constraints and Image Quality

Technical constraints in medical imaging, such as resolution differences, noise levels, and contrast variations, can significantly impact segmentation accuracy. Lower-resolution images may obscure fine structures, leading to inconsistencies in boundary delineation [?].

When combined with long sessions, images with insufficient quality might also increase the cognitive load of the annotators, leading to fatigue and reduced precision in labeling [?]. This is particularly relevant in histopathological studies, where the staining process and tissue preparation can introduce color variations and artifacts that affect image quality, even if the same scanning equipment is used [?].

1.2.3 Research Question

Given the challenges posed by inconsistent labels in medical image segmentation, this work aims to address the following research question:

Research Question

How can we develop a learning approach for ISS tasks in medical images that can adapt to inconsistent labels without requiring explicit supervision of labeler performance, while addressing challenges related to variability in expertise levels and technical constraints, and maintaining interpretability and generalization?

1.3 State of the Art

Certainly, in general Machine Learning (ML) classification tasks³ where multiple annotators are involved, Majority Voting (MV) is by far the simplest possible approach to implement. This concept was born multiple times and divergently in multiple fields, but it was described as relevant for ML and pattern recognition labeling for classification in [?], in which the approach is exposed as simple, yet powerful. The authors describe the MV as a method that can be used to improve

³In this work, image segmentation is considered as a particular case of classification in which target classes are assigned pixel-wise.

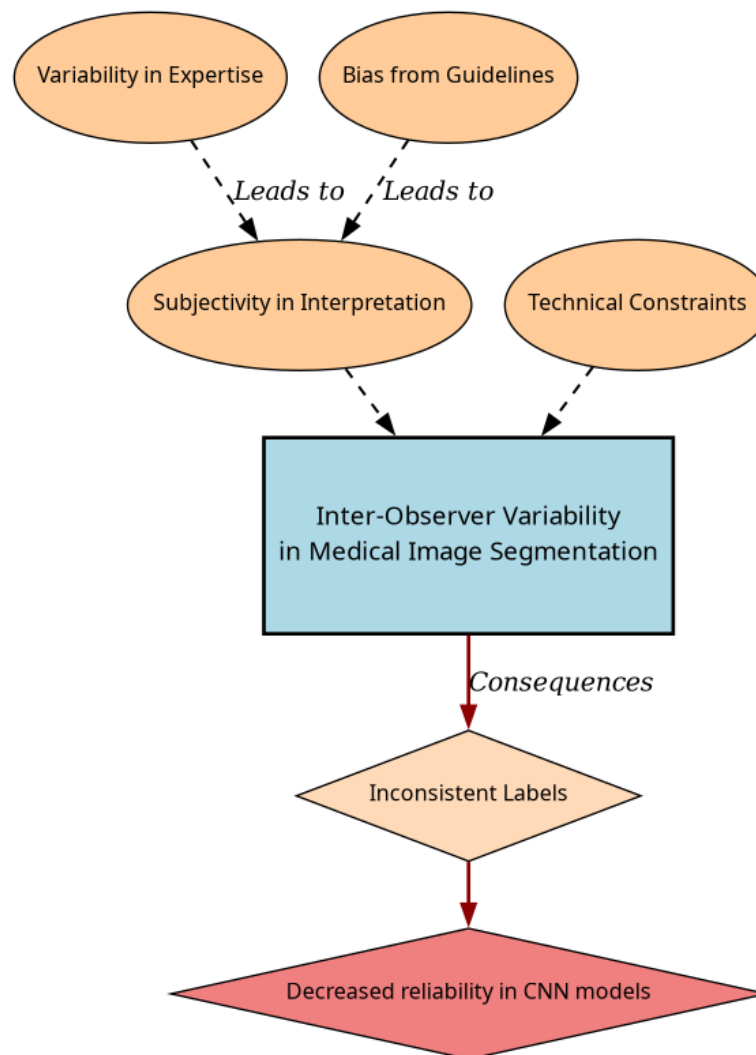


Figure 1-4 Summary diagram for problem Statement

the accuracy of classification tasks by combining the labels of multiple annotators. The method is based on the assumption that the majority vote of the annotators is more likely to be correct than the vote of a single annotator. The authors also describe the method as a straightforward way to improve the accuracy of classification tasks without the need for complex algorithms or additional data. The authors also prove this method to deliver very similar results to more complicated approaches (Bayesian, logistic regression, fuzzy integral, and neural network) in the particular task of Optical Character Recognition (OCR). Despite its simplicity, modern solutions for delivering accurate medical image segmentation models still rely on Majority Voting at some stage. In [?], a majority voting strategy is used for delivering a final output based on the labels of multiple models (VGG16-Segnet, Resnet-18 and Alexnet) in CT images for Liver Tumor Segmentation. Majority voting as a technique for label aggregation is a powerful approach for its simplicity in many use cases in which the target to be labeled is not tied to an expertise related task, otherwise, the assumption of equal expertise among the labelers may introduce bias in the final label, which is not desirable in the case of highly technical annotations like medical images. In subsection ??, a review is done on literature which no longer assumes the naive approach of equal expertise among labelers and face the challenge of learning from inconsistent labels will be presented.

1.3.1 Facing Annotation Variability in Medical images

Learning from crowds approaches in general face the challenge of not having a ground truth label and hence, an intrinsic difficulty in measuring the real reliability of the labelers annotations. Some approaches assume beforehand a certain level of expertise for each labeler based on experience as an input, like in [?], which introduce the concept of max margin majority voting, using the reliability vector as weights for the binary and multiclass classifier. The crowdsourcing margin is the minimal difference between the aggregated score of the potential true label and

the scores for other alternative labels. Accordingly, the annotators' reliability is estimated as generating the largest margin between the potential true labels and other alternatives. The problem introduced in this approach is assuming an stationary reliability per expert across the whole input space, which is imprecise since annotators performance may change between different tasks or even between different regions of the same image.

STAPLE Mechanism

The **Simultaneous Truth and Performance Level Estimation (STAPLE)** algorithm, introduced in [?] is a probabilistic framework that estimates a hidden true segmentation from multiple segmentations provided by different raters. It also estimates the reliability of each rater by computing their sensitivity and specificity.

The **STAPLE** algorithm's goal is to maximize the log likelihood function:

$$(\hat{\mathbf{p}}, \hat{\mathbf{q}}) = \arg \max_{\mathbf{p}, \mathbf{q}} \ln f(\mathbf{D}, \mathbf{T} \mid \mathbf{p}, \mathbf{q}). \quad (1-1)$$

Where $\mathbf{D}$ is the set of segmentations provided by the raters, $\mathbf{T}$ is the hidden true segmentation, p is the sensitivity and q is the specificity of the raters.

This is achieved by using the Expectation-Maximization algorithm to maximize the log likelihood function, which is done iteratively with step computations:

$$\begin{aligned} (p_j^{(k)}, q_j^{(k)}) = \arg \max_{p_j, q_j} & \sum_{i: D_{ij}=1} W_i^{(k-1)} \ln p_j \\ & + \sum_{i: D_{ij}=1} \left(1 - W_i^{(k-1)}\right) \ln(1 - q_j) \\ & + \sum_{i: D_{ij}=0} W_i^{(k-1)} \ln(1 - p_j) \\ & + \sum_{i: D_{ij}=0} \left(1 - W_i^{(k-1)}\right) \ln q_j. \end{aligned} \quad (1-2)$$

431 The weight terms $W_i^{(k-1)}$ and $(1 - W_i^{(k-1)})$ represent the current estimates of the
 432 probability that voxel i belongs or does not belong to the true segmentation from
 433 the previous E-step. The log-likelihood contributions $\ln p_j$, $\ln(1 - q_j)$, $\ln(1 - p_j)$, and
 434 $\ln q_j$ account for true positives, false positives, false negatives, and true negatives,
 435 respectively, weighted by the current estimate of true segmentation probabilities.

436 The capacity of STAPLE to accurately estimate the true segmentation, even in the
 437 presence of a majority of raters generating correlated errors, was demonstrated,
 438 which makes it theoretically a strong choice for setting a ground-truth in binary or
 439 multiclass medical ISS tasks.

440 The popularity and performance of STAPLE has led to its usage in modern
 441 applications medical image, 3d spatial images due to its assumption of decision
 442 space being based on voxel-wise decisions, like the authors in [?] which applied
 443 the algorithm on Positron Emission Tomography (PET) images. Other authors still
 444 rely heavily on STAPLE for setting a ground truth consensus for histopathological
 445 images, like [?].

446 However, the STAPLE algorithm has some limitations. It assumes independent
 447 rater errors, which may not hold in practice, leading to biased estimates. STAPLE
 448 is also sensitive to low-quality annotations, potentially degrading final
 449 segmentations if the weights are not initialized correctly. The algorithm tends to
 450 over-smooth results, blurring fine details, and struggles with multi-class
 451 segmentation. Computationally, it is expensive due to its iterative
 452 Expectation-Maximization (EM) approach. Additionally, STAPLE cannot correct
 453 systematic biases in annotations and depends on initial estimates, impacting
 454 accuracy. Lastly, the estimated performance levels lack interpretability, making it
 455 difficult to assess annotator reliability effectively.

456 Finally, while STAPLE proves valuable for aggregating existing labels and
 457 supporting other methodologies, its primary limitation lies in its inability to
 458 generate annotations for novel, unlabeled images. This restricts its applicability to
 459 scenarios where multiple annotations already exist, rather than serving as a
 460 standalone solution for new image segmentation tasks.

U-shaped CNNs

Since the introduction of U-Net [?] in 2015 for biomedical image segmentation, U-shaped CNNs have become a prevalent architecture in medical image segmentation tasks. The U-Net's success stems from its ability to capture both global and local information through its contracting and expanding paths, making it particularly effective for complex and heterogeneous structures, even with limited annotated data. This architecture has been successfully applied to various medical image segmentation tasks, including organ segmentation, tumor segmentation, and brain structure segmentation [?].

The U-Net architecture consists of a symmetric encoder-decoder structure with skip connections. The encoder path progressively reduces spatial dimensions while increasing feature channels through a series of convolutional and max-pooling layers, capturing high-level semantic information. The decoder path uses transposed convolutions to gradually recover spatial resolution while reducing feature channels. Skip connections between corresponding encoder and decoder layers preserve fine-grained details by concatenating high-resolution features from the encoder with upsampled features in the decoder, enabling precise localization of structures.

In [?] two networks are trained for delivering a final segmentation. One network is trained to estimate the annotators reliability and another one is trained to segment the image. The first network is a Deep Neural Network that takes as input features of image and the one-hot encoded identifiers of the labelers and outputs a reliability map across the image feature space. This map is then used to weight the contribution of each annotator to the final segmentation. The second network is the U-Net used for segmentation.

In this approach, it is assumed that the images are labeled for at least one labeler and not all of them, which is closer to a real world scenario, in which images with variability in the amount of annotations per patch is common. Hence, the input data can be modeled as:

$$\mathcal{D} = (\mathbf{X}, \tilde{\mathbf{Y}}) = \{(\mathbf{x}_n, \tilde{\mathbf{y}}_n^r) : n = 1, \dots, N; r \in R_n\}, \quad (1-3)$$

Where every $\mathbf{x}_n$ is an input patch from a ROI in one WSI, $\tilde{\mathbf{y}}_n$ is the noisy annotation from the r labeler, N is the number of patches in the dataset and $R_n \subset \{1, \dots, R\}$ is the set of labelers that annotated the image $\mathbf{x}_n$.

The authors then assume the annotator network to deliver a reliability map $\{\hat{\mathbf{A}}_\phi^{(r)}(\mathbf{x})\}_{r \in R_n}$ with different dimensions:

- **Crowdsourcing (CR)** global: a single reliability vector per labeler with dimensions C which represent global reliability of the labeler across all input space.
- **CR** image: a single reliability vector per image per labeler with dimensions C which represent local reliability of the labeler across the image.
- **CR** pixel: a reliability matrix per image per labeler, with dimensions C which represent local reliability of the labeler across all the pixels in the image.

These differences in dimensions are determined by the feature extraction space from segmentation network which feed the input of the annotator network, which the authors vary for experimentation purposes.

Let $\mathbf{p}_\theta(\mathbf{x}_n)$ be the estimated latent (ground truth) segmentation produced by the segmentation U-Net network. The estimated segmentation probability mask for each annotator is then computed as the product:

$$\mathbf{p}_{\theta, \phi}^{(r)}(\mathbf{x}_n) := \mathbf{A}_\phi^{(r)}(\mathbf{x}) \odot \mathbf{p}(\mathbf{x}_n), \quad (1-4)$$

where $\odot$ is the element-wise product and ϕ and θ are the parameters of the annotator network and the segmentation UNet network, respectively, being the latter initialized with a ResNet34 backbone pre-trained on ImageNet.

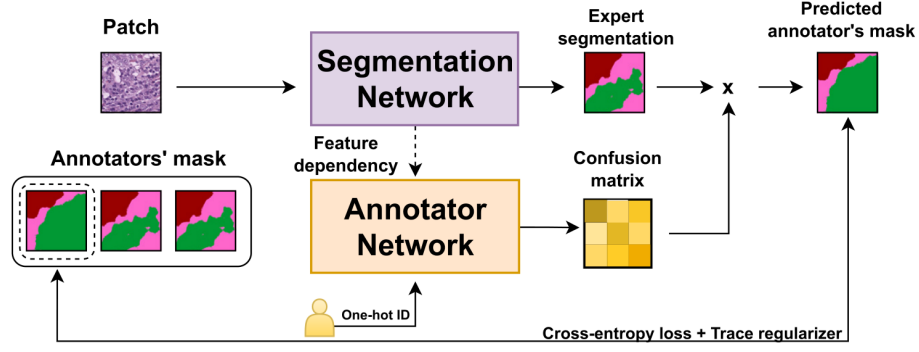


Figure 1-5 Proposed framework for the approach in [?].

511 The authors propose a loss function involving cross-entropy and a trace based
 512 regularization on the reliability map, originally proposed in [?] which combined,
 513 looks like:

$$\mathcal{L}(\theta, \phi) := \sum_{n=1}^N \sum_{r=1}^R \mathbb{I}(\tilde{\mathbf{y}}_n^{(r)} \in R_n) \cdot \left[\text{CE} \left(\mathbf{A}_{\phi}^{(r)}(\mathbf{x}_n) \cdot \mathbf{p}_{\theta}(\mathbf{x}_n), \tilde{\mathbf{y}}_n^{(r)} \right) + \lambda \cdot \text{tr} \left(\mathbf{A}_{\phi}^{(r)}(\mathbf{x}_n) \right) \right] \quad (1-5)$$

514 Being $\mathbb{I}$ the indicator function, CE the cross-entropy loss, and λ the regularization
 515 parameter.

516 When evaluated on a Triple Negative Breast Cancer dataset, this approach achieves
 517 a Dice coefficient of 0.7827, outperforming STAPLE (0.7039) and matching
 518 expert-supervised performance (0.7723). The CR image reliability modeling
 519 proved most effective, on the other hand CR pixel, while potentially offering
 520 finer-grained reliability estimation, requires significantly more training data.

521 Despite the proven performance of the approach by the authors, solving the problem
 522 of multiple labelers with two networks can be overwhelming for the optimization
 523 process, requiring large amounts of annotated data to properly codify the annotators
 524 spatial reliabilities, which could be managed by a single model with an appropriate
 525 loss function.

Bayesian Models

Bayesian approaches are a good choice for handling label noise and uncertainty in the labelers. In [?] the authors propose a novel approach based on Gaussian Processes for modeling the relationship between the annotators' reliability and the input data, while also preserving the interdependencies among the annotators. This is achieved by introducing **Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA)**, a framework based on the well known **Chained Gaussian Processes (CGP)**. CGP on itself cannot consider inter-annotator dependencies, thus, the authors introduce the **Correlated Chained Gaussian Processes (CCGP)** thus, the authors introduce the **CCGP** to model correlations between the GP latent functions, which are supposed to be generated from a **Semi-Parametric Latent Factor Model (SLFM)**:

$$f_j(\mathbf{x}_n) = \sum_{q=1}^Q w_{j,q} \mu_q(\mathbf{x}_n), \quad (1-6)$$

where $f_j : \mathcal{X} \rightarrow \mathbb{R}$ is a **Latent Function (LF)**, $\mu_q(\cdot) \sim \mathcal{GP}(0, k_q(\cdot, \cdot))$ with $k_q : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ being a kernel function, and $w_{j,q} \in \mathbb{R}$ is a combination coefficient ($Q \in \mathbb{N}$). This leads to a joint distribution of the form:

$$p(\mathbf{y}, \hat{\mathbf{f}}, u | \mathbf{X}) = p(\mathbf{y} | \boldsymbol{\theta}) \prod_{j=1}^J p(\mathbf{f}_j | \mathbf{u}) p(\mathbf{u}), \quad (1-7)$$

where $\mathbf{y}$ is the vector of noisy labels, $\hat{\mathbf{f}}$ is the vector of latent functions, u represents the inducing points, and $\mathbf{X}$ is the input data.

Combined with inducing-variables based methods for sparse GP approximations, and maximizing an **Evidence Lower Bound (ELBO)** for the estimation of the variational parameters, the authors reach a model whose variational expectations

are not analytically tractable, and hence, the authors derive a Gaussian-Hermite quadrature approach.

Finally, the authors extend this approach for being applied to classification and regression, reaching the only known approach to involve chained gaussian processes in multiple annotators classification and regression tasks while preserving the interdependencies among the annotators, and also outperforming GPC-MV⁴, MA-LFC-C⁵, MA-DGRL⁶, MA-GPC⁷, MA-GPCV⁸, MA-DL⁹, KAAR¹⁰.

CCGPMA on itself proposes a good approach for handling label noise and uncertainty in the labelers for regression and classification tasks, while also preserving the interdependencies among the annotators, however, it does not face the image segmentation problem, which is the main focus of this work. Besides, handling so many latent functions during the optimization process is computationally expensive, making it on itself infeasible for large and high resolution datasets.

1.3.2 Facing Noisy Annotations and Low-Quality Data

The problem of low-quality data and noisy annotations has been tackled with various strategies. One such approach is the use of deep learning models that incorporate loss functions designed to mitigate the effects of unreliable labels. Traditional methods such as MV or EM have been widely used for aggregating multiple annotators' inputs. However, they assume a homogeneous reliability of annotators, which may not hold in real-world scenarios.

⁴A GPC using the MV of the labels as the ground truth.

⁵A LRC with constant parameters across the input space.

⁶A multi-labeler approach that considers as latent variables the annotator performance.

⁷A multi-labeler GPC, which is an extension of MA-LFC.

⁸An extension of MA-GPC that includes variational inference and priors over the labelers' parameters.

⁹A Crowd Layer for DL, where the annotators' parameters are constant across the input space.

¹⁰A kernel-based approach that employs a convex combination of classifiers and codes labelers dependencies.

Loss Functions in Deep Learning Models

Loss functions are fundamental components in deep learning models that quantify how well a model's predictions match the ground truth. They serve as the objective function that guides the learning process by measuring the discrepancy between predicted and actual values. In classification tasks, the most common loss functions are Cross-Entropy (CE) and Mean Absolute Error (MAE) [?]. CE is particularly effective for classification as it heavily penalizes confident but wrong predictions, though it can be sensitive to noisy labels. MAE, on the other hand, is more robust to outliers and assigns equal weights to all mistakes, but typically requires more training iterations. For image segmentation tasks, specialized loss functions have been developed to handle the unique challenges of pixel-wise classification. The Dice loss, which measures the overlap between predicted and ground truth regions, is widely used in medical image segmentation [?]. More recently, the Generalized Cross Entropy (GCE) loss has emerged as a robust alternative that combines the benefits of both CE and MAE, allowing for better handling of noisy labels through a tunable parameter that controls sensitivity to outliers. However, to the best of our knowledge, none of the current loss functions approaches have focused on the multiple labelers scenario, neither on the per labeler spatial reliability estimation.

Generalized Cross-Entropy for Multiple Annotators Classification

A more recent approach was proposed by [?], introducing a **Generalized Cross-Entropy-based Chained Deep Learning (GCECDL)** framework. This method addresses the limitations of traditional label aggregation techniques by modeling each annotator's reliability as a function of the input data. The approach effectively mitigates the impact of noisy labels by using a noise-robust loss function, balancing Mean Absolute Error (MAE) and Categorical Cross-Entropy (CE). Unlike prior approaches, **GCECDL** accounts for the dependencies among annotators while encoding their non-stationary behavior across different data

595 samples. Their experiments on multiple datasets demonstrated superior
 596 predictive performance compared to state-of-the-art methods, particularly in cases
 597 where annotations were highly inconsistent.

598 The strategy of the authors effectively unlocks the potential of **ML** models to handle
 599 low-quality data and noisy annotations, but it is bounded to classifications tasks
 600 only, not being by itself applicable to segmentation tasks. The TGCE equation for
 601 handling multiple annotators is defined as:

$$\text{TGCE}(\mathbf{y}, f(\mathbf{x}); \tilde{\lambda}_x, \tilde{C}) = \tilde{\lambda}_x \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q} + (1 - \tilde{\lambda}_x) \frac{1 - (\tilde{C})^q}{q}, \quad (1-8)$$

602 where $\tilde{\lambda}_x$ represents the annotator reliability, $\tilde{C}$ is a constant, q is a parameter that
 603 controls the balance between MAE and CE behavior, $\mathbf{y}$ is the annotation vector, and
 604 $f(\mathbf{x})$ is the model prediction. This approach is more deeply discussed in chapter ??.

605 1.4 Aims

606 1.4.1 General Aim

607 The main purpose of this work is to adaptively infer the best possible segmentation
 608 in **ISS** tasks for medical images through a novel approach, without needing separate
 609 inputs about the labeler while offering an interpretable reliability space per labeler
 610 and properly facing the performance inconsistency across the annotators space and
 611 the variability of images quality.

Table 1-1 Summary of state-of-the-art approaches for handling multiple annotators in medical image segmentation

Approach	Key Characteristics	Advantages	Limitations
MV [?]	Simple aggregation of multiple annotators' labels	- Simple to implement - No complex algorithms required - Proven effective in many cases	- Assumes equal expertise - Sensitive to correlated errors - May introduce bias in expert tasks
STAPLE [?]	Probabilistic framework estimating true segmentation and rater reliability	- Handles binary and multiclass segmentation - Estimates rater sensitivity/specificity - Robust to correlated errors	- Computationally expensive - Sensitive to low-quality annotations - Tends to over-smooth results - Limited interpretability
U-Net based approaches [?]	Deep Learning architecture with encoder-decoder structure	- Captures both global and local information - Effective for complex structures - Works well with limited data	- Requires large amounts of training data - Complex optimization process - May need specialized loss functions
Bayesian Models (CCGPMA) [?]	Gaussian Processes modeling annotator reliability	- Preserves annotator interdependencies - Handles label noise effectively - Works for classification and regression	- Computationally expensive - Not directly applicable to segmentation - Complex optimization process
Generalized Entropy (GCE) [?]	Cross-Loss function balancing MAE and CE	- Robust to noisy labels - Accounts for non-stationary behavior - Handles annotator dependencies	- Limited to classification tasks - Requires parameter tuning - May need adaptation for segmentation

612 1.4.2 Specific Aims

- 613 • To develop a novel loss function for ISS tasks in medical images, which
614 accurately infers the underlying ground truth segmentation mask without
615 needing separate inputs about the labeler performance.
- 616 • To introduce a tensor map which codifies the reliability of each labeler based
617 only on the inputs, allowing the model to implicitly weigh the labelers based
618 on their performance across the mask and classes space.
- 619 • To develop and test a deep learning model for ISS tasks in medical images,
620 which can learn from inconsistent labels and improve the segmentation
621 performance compared to other solutions in state of the art.

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CHAPTER

TWO

626

CONCEPTUAL PRELIMINARIES

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2.1 Modern Concept of Digital Image

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A digital image is a numerical representation of a visual scene, captured through various imaging devices and stored in a computer. From a mathematical perspective, a digital image can be represented as a function $f(x, y)$ that maps spatial coordinates (x, y) to intensity values. In the discrete domain, this function is sampled at regular intervals, creating a matrix of values known as pixels (picture elements).

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2.1.1 Types of Digital Images

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Grayscale Images

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Grayscale images are the simplest form of digital images, where each pixel represents a single intensity value. Mathematically, a grayscale image can be represented as a 2D matrix I of size $M \times N$, where each element $I(i, j)$ represents the intensity at position (i, j) . The intensity values typically range from 0 (black) to 255 (white) in 8-bit images, or from 0 to 65535 in 16-bit images.

640 **Color images**

641 Color images extend the grayscale concept by representing each pixel with multiple
642 channels, typically Red, Green, and Blue (RGB). A color image can be represented
643 as a 3D matrix I of size $M \times N \times 3$, where $I(i, j, k)$ represents the intensity of the
644 k -th color channel at position (i, j) . Other color spaces like HSV (Hue, Saturation,
645 Value) or CMYK (Cyan, Magenta, Yellow, Key) are also commonly used in different
646 applications.

647 **Multispectral images**

648 Multispectral images capture information across multiple wavelength bands
649 beyond the visible spectrum. These images can be represented as a 3D matrix I of
650 size $M \times N \times B$, where B is the number of spectral bands. Each band $I(i, j, b)$
651 represents the intensity at position (i, j) for the b -th spectral band. This
652 representation is particularly useful in medical imaging, remote sensing, and
653 several scientific applications.

654 **3D Images and Volumetric Data**

655 Three-dimensional images extend the concept of pixels to voxels (volume elements).
656 A 3D image can be represented as a 3D matrix V of size $M \times N \times D$, where D
657 represents the depth dimension. Each voxel $V(i, j, k)$ represents the intensity at
658 position (i, j, k) in the 3D space. This representation is fundamental in medical
659 imaging (CT, MRI), scientific visualization, and computer graphics.

660 **2.1.2 Mathematical Representations**

661 The mathematical foundation of digital images relies on several key concepts:

- 662 • **Sampling:** The process of converting a continuous image into a discrete
663 representation. According to the Nyquist-Shannon sampling theorem, the
664 sampling frequency must be at least twice the highest frequency present in
665 the image to avoid aliasing.
- 666 • **Quantization:** The process of converting continuous intensity values into
667 discrete levels. The number of quantization levels determines the image's
668 bit depth and affects its quality and storage requirements.
- 669 • **Resolution:** The number of pixels per unit length in an image, typically
670 measured in pixels per inch (PPI) or dots per inch (DPI).
- 671 • **Dynamic range:** The ratio between the maximum and minimum measurable
672 light intensities in an image, often expressed in decibels (dB).

673 The mathematical representation of a digital image can be expressed as:

$$I(x, y) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} f(i, j) \cdot \delta(x - i, y - j) \quad (2-1)$$

674 where $I(x, y)$ is the digital image, $f(i, j)$ represents the intensity values, and $\delta(x -$
675 $i, y - j)$ is the Kronecker delta function [?].

676 For color images, the representation extends to:

$$I(x, y) = \begin{bmatrix} I_R(x, y) \\ I_G(x, y) \\ I_B(x, y) \end{bmatrix} \quad (2-2)$$

677 where I_R , I_G , and I_B represent the red, green, and blue channels respectively.

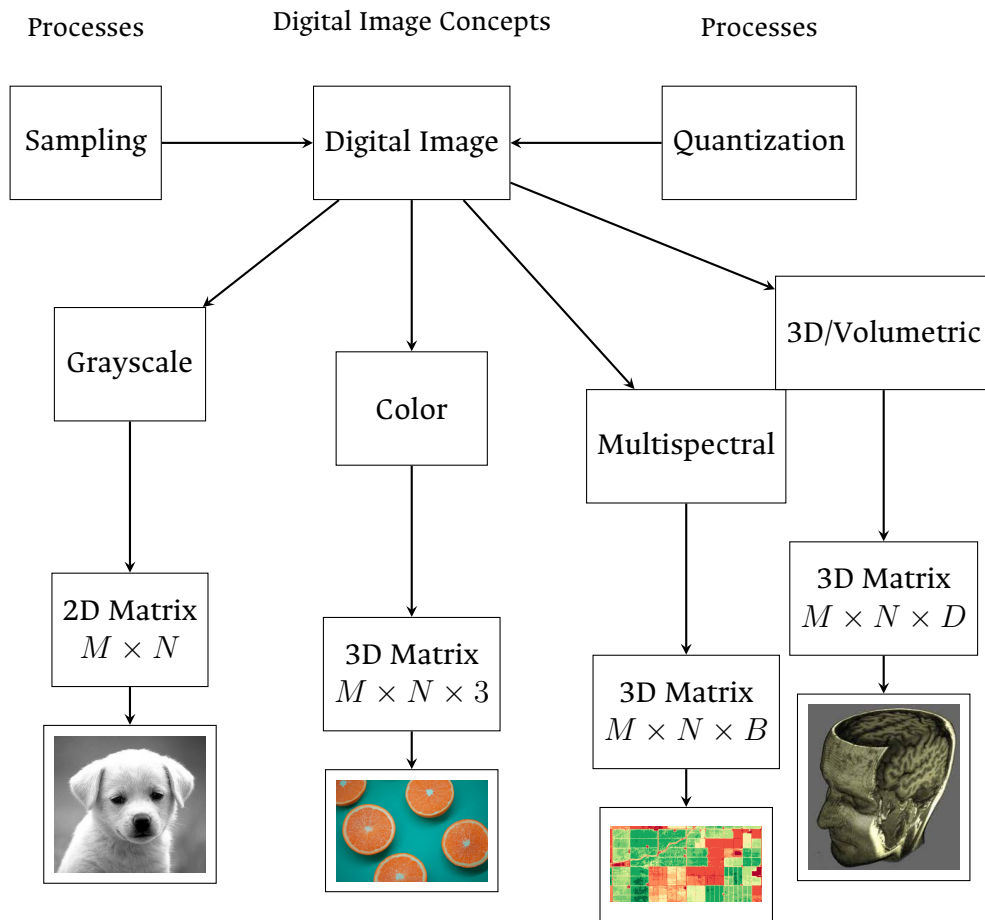


Figure 2-1 Overview of digital image concepts and their mathematical representations. The figure shows the main types of digital images (grayscale, color, multispectral, and volumetric), their mathematical representations, and the fundamental processes of sampling and quantization. Example images are included to illustrate each type.

2.2 Digital Histopathological Images

Digital histopathology represents a significant advancement in medical imaging, where traditional glass slides containing tissue samples are digitized using specialized scanning devices. This transformation has revolutionized the field of pathology by enabling remote diagnosis, computer-aided analysis, and digital archiving of tissue samples [?]. This process has evolved significantly over the past few decades, as shown in Figure ??.

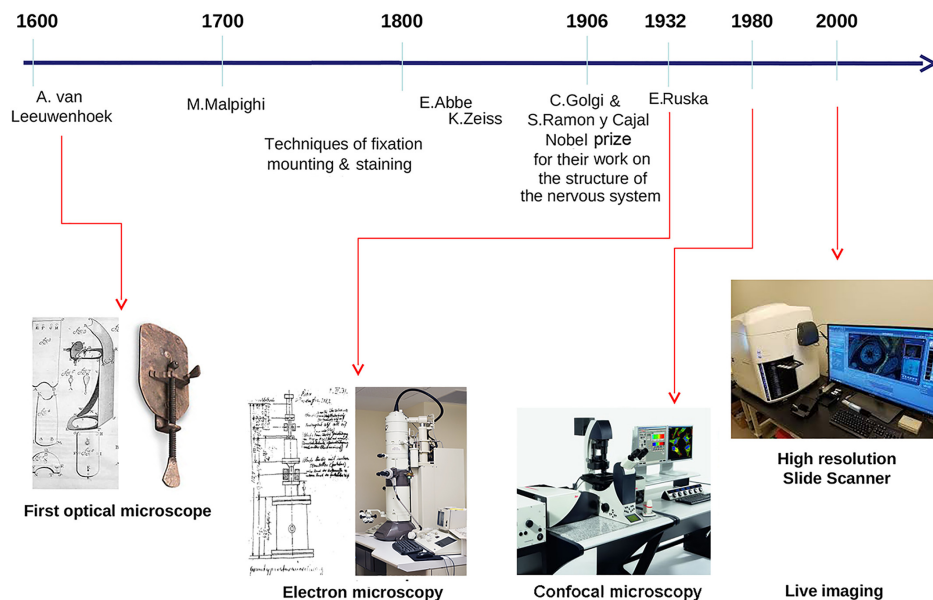


Figure 2-2 Histology evolution timeline. (Image from [?]).

2.2.1 Whole Slide Imaging (WSI)

Whole Slide Imaging (WSI) is the process of digitizing entire glass slides at high resolution, creating a digital representation that can be viewed, analyzed, and shared electronically. Modern **WSI** scanners use sophisticated optical systems that capture multiple fields of view at high magnification, which are then stitched

690 together to create a seamless digital image [?]. These systems incorporate
691 high-resolution objectives with magnifications ranging from 20x to 40x, precise
692 motorized stages for accurate slide positioning, automated focus systems to
693 maintain image quality, and high-quality cameras equipped with large sensor
694 arrays. The resulting digital slides can reach sizes of several gigabytes, containing
695 billions of pixels that capture the microscopic details of tissue samples [?]. Figure
696 ?? shows a whole slide imaging system by Omnyx for slide digitization and a
697 comprehensive digital pathology interface from Omnyx designed to streamline
698 pathologists' diagnostic workflow [?].



Figure 2-3 (Above) Whole slide imaging system by Omnyx for slide digitization. (Below) Comprehensive digital pathology scan visualization from Omnyx designed to streamline pathologists' diagnostic workflow. (From [?]).

699 2.2.2 Regions of Interest (ROI)

700 In digital histopathology, **Region of Interests (ROIs)** are specific areas within a
701 whole slide image that contain diagnostically relevant information. The
702 importance of **ROIs** lies in their ability to focus computational analysis on relevant
703 areas, reduce computational complexity in automated systems, facilitate targeted
704 diagnosis and research, and enable efficient storage and transmission of critical
705 information.

706 Selecting **ROIs** is a fundamental step in medical image analysis across all imaging
707 modalities, where an ROI represents a subset of an image appropriate for the
708 intended analysis and is typically identified manually by experts [?]. In modern
709 pathology, the analysis involves processing multidimensional and high-resolution
710 **WSI** tiles automatically, containing an overwhelming quantity of structural and
711 functional information. Despite recent improvements in computing capacity,
712 analyzing such extensive data remains challenging but vital for accurate analysis.
713 Automatic ROI detection can significantly reduce the number of pixels to be
714 processed, speed up the analysis, improve accuracy, and reduce dependency on
715 pathologists [?].

716 The challenge of acquiring image domain annotations for ROI detection is
717 particularly significant due to the knowledge and commitment required of expert
718 pathologists. Pathologists often identify regions in whole slide images with
719 diagnostic relevance rather than examining the entire slide, with a positive
720 correlation between the time spent on these critical image regions and diagnostic
721 accuracy [?]. This selective viewing behavior highlights the importance of
722 developing automated systems that can replicate pathologists' diagnostic patterns.

723 Recent advances in deep learning techniques have offered improvements in
724 computer-aided diagnosis systems for ROI detection. Innovative approaches
725 include generating heatmaps to represent pathologists' viewing patterns during
726 diagnosis and using these to guide deep learning architectures during training [?].

Such systems can integrate pathologists' domain expertise to enhance region of interest detection without needing individual case annotations, potentially surpassing traditional approaches based on color and texture image characteristics.

The clinical application of automated ROI detection has been demonstrated in specific diagnostic contexts, such as Human Epidermal Growth Factor Receptor 2 (HER2) grading for breast cancer patients. Existing HER2 grading relies on manual ROI selection, which is tedious, time-consuming, and suffers from inter-observer and intra-observer variability [?]. Studies have found that HER2 grade changes with ROI selection, emphasizing the critical importance of accurate and consistent ROI detection methods. Advanced approaches using Vision Transformer architectures have shown promising results, achieving high accuracy rates for ROI detection while significantly reducing processing time and improving diagnostic agreement with clinical scores [?].

These automated ROI detection methods not only enhance diagnostic accuracy and efficiency but also streamline annotation processes and aid in the training of new pathologists. By incorporating pathologists' expertise through heatmap generation and advanced deep learning architectures, these systems demonstrate the potential to revolutionize digital pathology workflows while maintaining the quality and reliability of diagnostic assessments [?] (Figure ??).

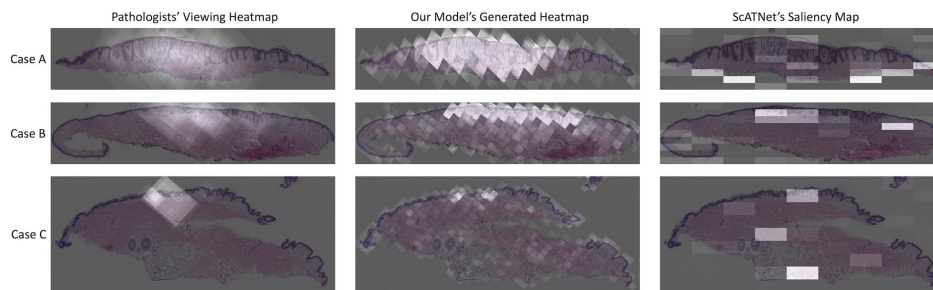


Figure 2-4 ROI heatmaps measured experimentally from pathologists view patterns and estimated heatmap from a U-Net model from [?].

746 2.2.3 Staining Techniques

747 Histopathological analysis relies heavily on various staining techniques to enhance
748 the visibility of different tissue components and cellular structures. The choice of
749 staining method depends on the specific diagnostic requirements and the type of
750 tissue being examined.

751 Hematoxylin and Eosin (H&E)

752 Hematoxylin and Eosin (H&E) staining is the most widely used technique in
753 histopathology, particularly in breast cancer diagnosis [?]. This staining method
754 provides essential visualization through two components: hematoxylin, which
755 stains cell nuclei blue/purple to highlight nuclear morphology, and eosin, which
756 stains cytoplasm and extracellular matrix pink/red to reveal tissue architecture.

757 The popularity of H&E staining in breast cancer histopathology stems from its
758 ability to clearly visualize tumor architecture and growth patterns, distinguish
759 between different types of breast cancer, identify important diagnostic features
760 like nuclear pleomorphism, and assess tumor grade and stage. Beyond breast
761 cancer, H&E staining finds extensive application across various medical specialties
762 including general pathology, dermatology, gastroenterology, neurology, and
763 oncology.

764 Special Stains

765 In addition to H&E, various special stains are used for specific diagnostic purposes.
766 Immunohistochemistry (IHC) uses antibodies to detect specific proteins, playing a
767 crucial role in subtyping breast cancers. Key IHC stains include Estrogen Receptor
768 (ER) staining for detecting estrogen receptors, Progesterone Receptor (PGR)
769 staining for assessing progesterone receptor status, Human Epidermal Growth
770 Factor Receptor 2 (HER2) staining for evaluating HER2 protein expression, and

771 Ki67 staining for measuring cellular proliferation rates. These markers are
772 particularly crucial in breast cancer diagnosis and treatment planning, as they help
773 determine the molecular subtype of the cancer and guide personalized therapeutic
774 approaches. Other specialized stains include Periodic Acid-Schiff (PAS) for
775 highlighting carbohydrates and basement membranes, Masson's Trichrome for
776 distinguishing between collagen and muscle fibers, and silver stains for detecting
777 microorganisms and nerve fibers. These specialized staining techniques
778 complement H&E by providing additional diagnostic information that is crucial for
779 accurate diagnosis and treatment planning [?]. Examples of these staining
780 techniques are shown in Figure ??.

781 2.3 Deep Learning Fundamentals

782 Deep Learning has emerged as a powerful subset of Machine Learning,
783 revolutionizing the field of artificial intelligence. Its roots can be traced back to the
784 early development of artificial neural networks in the 1940s and 1950s, with
785 significant milestones including the perceptron in 1958 and the backpropagation
786 algorithm in the 1980s. However, it wasn't until the early 21st century, with the
787 advent of more powerful computational resources and the availability of large
788 datasets, that deep learning truly began to flourish [?].

789 2.3.1 Learning Paradigms

790 Machine Learning systems can be categorized into three main learning paradigms.
791 The most common approach is supervised learning, where models learn from
792 labeled data by mapping inputs to known outputs. This paradigm requires a large
793 amount of labeled training data, which can be expensive and time-consuming to
794 acquire. Semi-supervised learning offers a hybrid approach that leverages both

labeled and unlabeled data [?], proving particularly useful when labeled data is scarce but unlabeled data is abundant. Finally, unsupervised learning enables models to discover patterns and structures from unlabeled data without explicit guidance [?], making it valuable for tasks like clustering and dimensionality reduction. Being deep learning a subset of Machine Learning, it is based on the same paradigms. An illustration of the three main learning paradigms is shown in Figure ??.

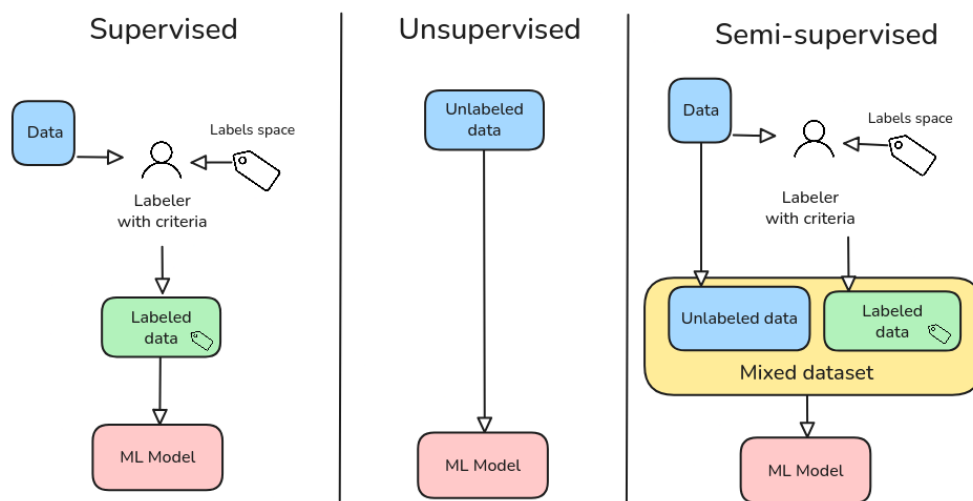


Figure 2-5 Common learning paradigms.

2.3.2 Architecture and Training

Deep Learning architectures are characterized by their layered structure, where each layer progressively extracts and transforms features from the input data [?]. The early layers typically focus on low-level feature extraction, such as edges, textures, and basic patterns in the case of image processing [?]. As information flows through the network, middle layers combine these basic features into more complex representations. The final layers perform high-level reasoning and make the ultimate predictions or classifications.

The training process commonly employs gradient descent as its primary optimization strategy, which iteratively adjusts the model's parameters to minimize a loss function. However, other optimization algorithms like Adam, RMSprop, and evolutionary strategies [?] [?], can also be effective for training deep networks. The loss function serves as a crucial component of the learning process in most optimization techniques, quantifying how well the model's predictions match the actual targets. By providing a measure of the model's performance, the loss function guides the chosen optimization process, enabling the network to learn meaningful patterns from the training data.

2.3.3 Challenges and Solutions

Despite their power, deep learning systems face several significant challenges. One of the most prominent issues is overfitting, where models may memorize training data instead of learning generalizable patterns [?]. This challenge is typically addressed through various regularization techniques such as dropout, L1/L2 regularization, and early stopping [?]. Another critical challenge is the substantial data requirements; deep learning models often need massive amounts of training data to achieve good performance [?], which can be a limiting factor in many applications. Additionally, the complex, layered nature of deep learning models makes them difficult to interpret, often referred to as "black boxes". This lack of transparency can be particularly problematic in critical applications where understanding the decision-making process is essential.

2.3.4 Deep Learning Frameworks

The development of powerful open-source frameworks has significantly accelerated deep learning research and applications. TensorFlow, developed by Google, provides a comprehensive ecosystem for building and deploying Machine

Learning models [?]. PyTorch, created by Facebook’s AI Research lab, offers dynamic computation graphs and has become particularly popular in research settings [?]. Caffe, known for its speed and modularity, is widely used in computer vision applications [?].

These frameworks have democratized deep learning by providing efficient implementations of common operations, automatic differentiation for gradient computation, and GPU acceleration for faster training. They also offer pre-trained models and transfer learning capabilities, along with active communities for support and knowledge sharing. The combination of these frameworks with modern hardware has enabled researchers and practitioners to develop increasingly sophisticated models, pushing the boundaries of what’s possible in artificial intelligence. TensorFlow and PyTorch have emerged as the two most prominent frameworks in the deep learning landscape, as shown in Figure ??, which presents data from Google Trends over the last five years (as of April 2025).

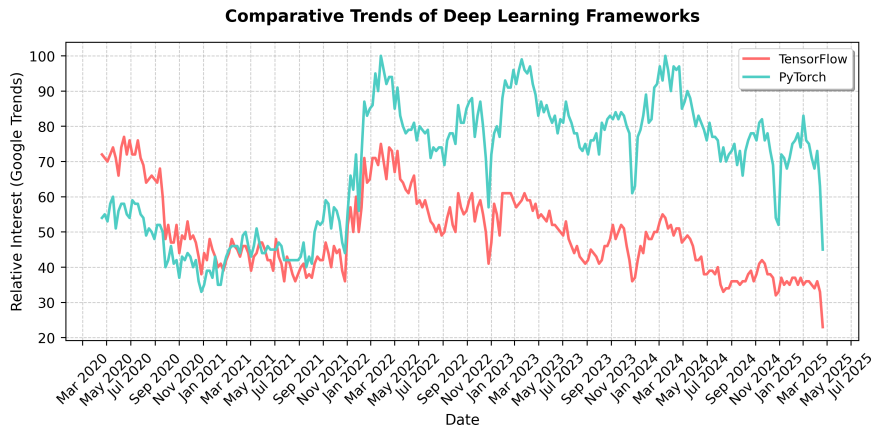


Figure 2-6 Comparative Trends of the top two most popular Deep Learning Frameworks.

2.4 Datasets and Data Sources

Throughout the development of this work, multiple datasets were used for evaluation of ISS models. The common elements of all these datasets are that they

contain RGB images and are crowdsourced with multiple labelers, where not necessarily all labeler label all images.

As it has been mentioned in Chapter ??, the main goal of this work is mainly focused on crowdsourced histopathology images semantic segmentation, however, these datasets present the following challenges:

- Distribution of segmentation labels is not uniform across the image, since some tissues and structures are more common than others.
- Visualization of performance of the models in debug time (like per epoch analysis) is not simple for non experts in the subject, which makes it hard to evaluate whether the model is overfitting or not at a glance.

For these reasons, multiple datasets were created this work in the pursuit of an initial evaluation of performance of the models against more traditional and familiar images before the focus on histopathology images. Once a satisfactory performance was achieved, the focus was shifted to histopathology images and further tunings on the models were performed if needed.

In any case, both the emulated noisy annotations datasets and the histopathology datasets somehow contained ground truth aggregation, either from the original source (in the case of emulated noisy annotations), the expert annotation (if available) or from the aggregation of multiple labelers ¹.

2.4.1 Oxford-IIIT Pet Dataset

Using the techniques described later in Section ??, the Oxford-IIIT Pet Dataset [?] was used to create a dataset with emulated noisy annotations. This dataset is officially available in the ‘tensorflow_datasets’ package, making it convenient to use in experiments. It is composed of 7349 images of pets, being 2371 of these images cats, and the rest 4978, dogs. Breed labels are included as well as segmentation masks for the classes: ‘background’, ‘border’, and ‘pet’. Figure ?? shows an example of the annotations in the original dataset.

¹STAPLE in the case of histopathology datasets with no expert annotations available

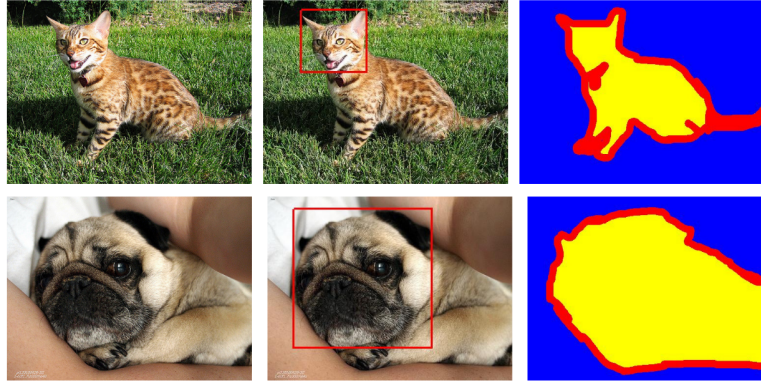


Figure 2-7 Annotations in the Oxford-IIIT Pet data. From left to right: pet image, head bounding box, and trimap segmentation (blue: background region; red: ambiguous region; yellow: foreground region).

2.4.2 Real histopathology datasets

Multi-Stain Breast Cancer Histological Dataset

The Multi-Stain Breast Cancer Histological Dataset [?] represents one of the largest publicly available collections of whole slide images (WSIs) from surgical resection specimens of primary breast cancer patients. This dataset is particularly valuable for our work because it contains matched pairs of H&E and IHC-stained tissue sections from the same tumor, with a total of 4,212 WSIs from 1,153 patients. The IHC stains include important biomarkers such as ER (Estrogen Receptor), PGR (Progesterone Receptor), HER2 (Human Epidermal Growth Factor Receptor 2), and KI67 (Ki67 Proliferation Marker), which are routinely used in breast cancer diagnosis and treatment planning (more on staining techniques in Section ??).

The dataset's relevance to our work stems from several key aspects. The matched H&E and IHC stains allows for studying the consistency of segmentation across different staining modalities, which is crucial for understanding how different visualization methods affect annotation quality. With 1,153 patients, the dataset provides a robust foundation for training and evaluating segmentation models in a real-world clinical setting. The inclusion of routine diagnostic cases makes the

dataset representative of actual clinical practice, where variations in staining quality and tissue preparation are common. Furthermore, the multiple biomarker stains (ER, PGR, HER2, KI67) enable to study how different tissue characteristics affect segmentation performance and annotator agreement.

This dataset serves as an ideal testbed for our crowdsourced segmentation approach. It allows us to evaluate how different staining modalities affect annotator performance and agreement, while also providing insights into the relationship between tissue characteristics and segmentation difficulty. The dataset's comprehensive nature enables validation of our models' performance across different biomarker expressions and assessment of the generalizability of our approach to real-world clinical data.

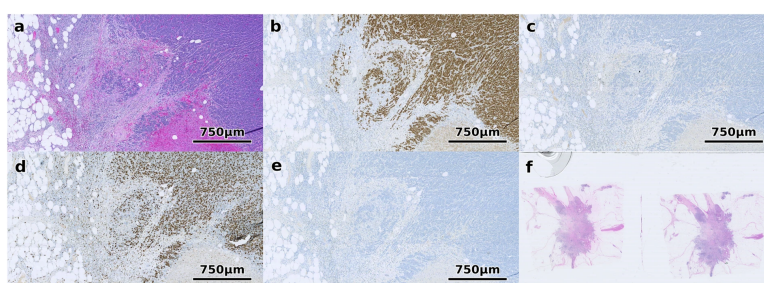


Figure 2-8 Different staining techniques obtained from multi-stain breast cancer dataset [?]. (a) shows H&E, (b) ER, (c) HER2, (d) Ki67 and (e) PGR. (f) shows an example of a WSI that was excluded since it contains multiple tissue sections.

Structured Crowdsourcing Dataset for Histology Images

The dataset presented by [?] is particularly relevant to our work as it represents one of the first systematic studies of crowdsourced annotations in histopathology. The authors recruited 25 participants with varying levels of expertise (from senior pathologists to medical students) to delineate tissue regions in 151 breast cancer slides using the Digital Slide Archive platform, resulting in over 20,000 annotated tissue regions. The annotation efforts are focused on Triple-Negative Breast Cancer (TNBC), an aggressive genomic subtype that comprises 15% of breast cancer cases [?].

916 Key aspects of this dataset make it valuable for our work. The systematic
917 evaluation of inter-participant discordance revealed varying levels of agreement
918 across different tissue classes, with low discordance for tumor and stroma, and
919 higher discordance for more subjectively defined or rare tissue classes. The
920 inclusion of feedback from senior participants helped in curating high-quality
921 annotations, demonstrating that fully convolutional networks trained on these
922 crowdsourced annotations can achieve high accuracy (mean AUC=0.945). The
923 dataset also provides evidence that the scale of annotation data significantly
924 improves image classification accuracy.

925 This dataset is particularly valuable for our work because it provides crucial
926 insights into how annotator expertise affects segmentation quality. It
927 demonstrates the feasibility of using crowdsourced annotations for training
928 accurate segmentation models, showing that even with varying levels of expertise,
929 aggregated annotations can produce reliable ground truth. The dataset includes a
930 systematic analysis of inter-annotator agreement, which is crucial for
931 understanding the challenges in crowdsourced histopathology segmentation and
932 informing the development of more robust segmentation approaches.

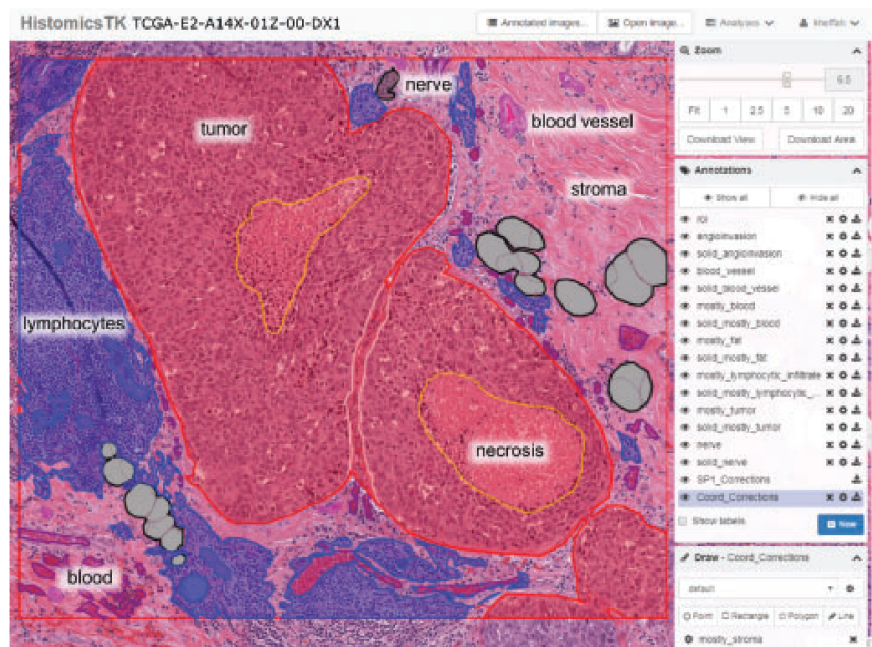


Figure 2-9 Screenshot of the DSA and HistomicsTK web interface while creating the crowdsourced annotations for the dataset presented by [?].

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CHAPTER

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THREE

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CHAINED GAUSSIAN PROCESSES

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3.1 Background and Related Methods

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In recent years, several approaches have been proposed to handle supervised learning problems in the context of multiple annotators [?] [?]. This section reviews key methods that form the foundation for understanding chained Gaussian processes and their application to multiple annotator scenarios in regression and classification tasks.

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3.1.1 Kernel Alignment-Based Annotator Relevance Analysis

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The Kernel Alignment-Based Annotator Relevance Analysis (KAAR) method, introduced in [?], addresses the challenge of estimating annotator expertise in scenarios where the ground truth is unavailable. The key innovation of KAAR lies

949 in its use of **Centered Kernel Alignment (CKA)** to measure the similarity between
 950 input features and annotator labels.

951 Given a dataset with input features $\mathbf{X}$ and labels from multiple annotators $\mathbf{Y}$, **KAAR**
 952 computes a kernel matrix $\mathbf{K}_\nu$ as a convex combination of R basis kernels:

$$\mathbf{K}_\nu = \sum_{r=1}^R \nu_r \mathbf{K}_r, \quad (3-1)$$

953 where $\mathbf{K}_r$ is computed using a kernel function over the input features, and ν_r
 954 represents the relevance weight for the r -th annotator. The weights are optimized
 955 by maximizing the CKA between $\mathbf{K}_\nu$ and a target kernel $\mathbf{F}$ computed from the
 956 input features:

$$\rho(\mathbf{K}_\nu, \mathbf{F}) = \frac{\langle \bar{\mathbf{K}}_\nu, \bar{\mathbf{F}} \rangle_F}{\|\bar{\mathbf{K}}_\nu\|_F \|\bar{\mathbf{F}}\|_F}, \quad (3-2)$$

957 A key advantage of **KAAR** is its ability to capture dependencies between annotators
 958 through the kernel framework, without requiring explicit parametric modeling of
 959 these relationships.

960 **3.1.2 Localized Kernel Alignment-Based Annotator** 961 **Relevance Analysis**

962 In [?] the authors extend **KAAR** by introducing locality in the kernel alignment
 963 framework. Rather than assuming constant annotator performance across the
 964 input space, **Localized Kernel Alignment-Based Annotator Relevance Analysis**
 965 (**LKAAR**) models the annotator relevance as a function of the input features. The
 966 kernel combination in **LKAAR** takes the form:

$$\mathbf{K}_q = \sum_{r=1}^R \mathbf{Q}_r \mathbf{K}_r \mathbf{Q}_r, \quad (3-3)$$

where $\mathbf{Q}_r$ is a diagonal matrix whose elements represent the local relevance of annotator r . These elements are computed as:

$$q_r(\mathbf{x}_n) = \begin{cases} \beta_0^{(r)} + \sum_{n'=1}^N \beta_{n'}^{(r)} \kappa_\beta(\mathbf{x}_n, \mathbf{x}_{n'}), & \text{if } n \in N_r \\ 0, & \text{Otherwise} \end{cases} \quad (3-4)$$

This localized approach allows **LKAAR** to better model inconsistent annotators whose performance varies across different regions of the input space [?].

3.1.3 Chained Gaussian Processes Model

The **CGP** model provides a probabilistic framework for modeling multiple outputs through a chain of Gaussian processes [?]. In the context of multiple annotators, **CGP** can be used to model both the ground truth and annotator behavior. The key idea is to model a likelihood function with J parameters as in [?]:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \prod_{n=1}^N p(y_n | \theta_1(\mathbf{x}_n), \dots, \theta_J(\mathbf{x}_n)), \quad (3-5)$$

where each parameter $\theta_j(\mathbf{x})$ is modeled using a separate Gaussian process. This framework provides remarkable flexibility in modeling complex relationships within the data [?]. The use of Gaussian processes enables natural uncertainty quantification through posterior distributions, allowing the model to express varying degrees of confidence in its predictions. The framework inherently handles missing data through its probabilistic nature, and the automatic relevance

determination property of GPs helps identify the most important features for prediction [?].

The CGP framework becomes particularly powerful when integrated with concepts from KAAR and LKAAR. By incorporating kernel alignment principles, the model can make more informed choices about kernel functions, better capturing the underlying structure of the data [?]. The framework can be extended to include input-dependent lengthscales, allowing for varying degrees of smoothness across the input space [?]. Furthermore, through multi-output Gaussian Process (GP) formulations, the model can capture complex dependencies capture complex dependencies between different annotators, providing a rich representation of how different experts' opinions relate to each other.

However, implementing CGPs in practice presents several significant challenges. The computational complexity of the model scales cubically with the dataset size, making it potentially prohibitive for large-scale applications [?]. The need for approximate inference techniques in non-Gaussian likelihood scenarios can introduce additional complexity and potential approximation errors [?]. When dealing with high-dimensional inputs, the model may struggle with the curse of dimensionality, requiring careful feature selection or dimensionality reduction preprocessing [?]. Additionally, the selection of appropriate kernel functions remains a critical challenge, as it significantly impacts the model's performance and its ability to capture relevant patterns in the data [?].

3.1.4 Regularized Chained Deep Neural Network

In [?], the authors propose a deep learning approach to multiple annotator learning, inspired by the chained Gaussian processes model. The method models both the ground truth estimation and annotator reliability through a deep neural network architecture. The likelihood function in Regularized Chained Deep Neural Network (RCDNN) takes the form:

$$p(\mathbf{Y}|\boldsymbol{\theta}) = \prod_{n=1}^N \prod_{r \in R_n} \left(\prod_{k=1}^K \zeta_{n,k}^{\delta(y_n^{(r)} - k)} \right)^{\lambda_n^{(r)}} \left(\frac{1}{K} \right)^{1 - \lambda_n^{(r)}}, \quad (3-6)$$

1009 where $\lambda_n^{(r)}$ represents the reliability of annotator r for instance n , and $\zeta_{n,k}$ is the
 1010 estimated probability of class k for instance n .

1011 The **RCDNN** architecture is built upon a sophisticated multi-layer structure that
 1012 combines several key innovations in deep learning [?]. At its core, it employs
 1013 multiple dense layers with carefully chosen activation functions, each designed to
 1014 capture different aspects of the annotator-data relationship. To prevent
 1015 overfitting, a comprehensive regularization strategy is implemented, combining
 1016 both L1 and L2 norms with dropout mechanisms [?]. The network architecture
 1017 splits into specialized branches: one dedicated to estimating the ground truth
 1018 labels, and another focused on modeling annotator reliability. This dual-branch
 1019 approach allows the network to simultaneously learn both the true underlying
 1020 patterns in the data and the individual characteristics of each annotator.
 1021 Furthermore, the implementation of Monte Carlo dropout during prediction
 1022 provides a mechanism for uncertainty estimation, offering not just predictions but
 1023 also confidence measures for those predictions.

1024 3.2 Tooling and Implementation

1025 The implementation of Chained Gaussian Processes for multiple annotator
 1026 learning requires robust and flexible software tools that can handle both the
 1027 probabilistic nature of Gaussian processes and the complexity of deep
 1028 architectures [?]. In this work, two key frameworks are leveraged: GPflow and
 1029 GPflux. These tools provide the foundation for building scalable and efficient
 1030 implementations of our models while maintaining the theoretical rigor necessary
 1031 for proper uncertainty quantification.

1032 The choice of these frameworks is motivated by several factors. First, both are
1033 built on top of TensorFlow, providing access to automatic differentiation and GPU
1034 acceleration capabilities. Second, they offer a modular design that allows for easy
1035 extension and modification, which is crucial when implementing novel model
1036 architectures. Third, they provide robust implementations of various Gaussian
1037 process models and inference methods, allowing us to focus on the specific
1038 challenges of multiple annotator learning rather than reimplementing basic GP
1039 functionality.

1040 3.2.1 GPflow Overview

1041 GPflow is a Python package for building Gaussian process models in TensorFlow.
1042 Developed as a modern and scalable alternative to GPy, it leverages TensorFlow's
1043 computational capabilities to provide efficient implementation of Gaussian process
1044 models. The framework is designed with four key principles in mind: ease of use,
1045 computational efficiency, extensibility, and automatic differentiation. This work
1046 was originally published in [?].

1047 At its core, GPflow provides implementations of various Gaussian process models
1048 including Gaussian Process Regression (GPR), Gaussian Process Monte Carlo
1049 (GPMC), Sparse Gaussian Process Regression (SGPR), and Sparse Gaussian Process
1050 Monte Carlo (SGPMC) [?]. These implementations are built on top of TensorFlow's
1051 computational graph framework, allowing for automatic differentiation and GPU
1052 acceleration where available.

1053 The framework's architecture is modular, with clear separation between model
1054 components such as kernels, likelihoods, and optimization routines. This
1055 modularity makes it straightforward to extend existing models or create new ones.
1056 GPflow includes a comprehensive suite of kernel functions, ranging from the basic
1057 RBF (Radial Basis Function) kernel to more sophisticated ones like the Spectral
1058 Mixture kernel.

1059 One of GPflow’s distinguishing features is its robust handling of model parameters.
1060 It provides facilities for parameter transformation (ensuring parameters remain in
1061 valid ranges during optimization), prior specification, and flexible optimization
1062 strategies. The framework also includes utilities for model evaluation and
1063 prediction, with built-in support for uncertainty estimation.

1064 3.2.2 GPflux Framework

1065 GPflux extends GPflow to enable the construction and training of Deep Gaussian
1066 Process (DGP) models using modern deep learning practices. Built on top of both
1067 GPflow and TensorFlow, GPflux provides a flexible framework for implementing
1068 deep probabilistic models that combine the expressiveness of deep neural
1069 networks with the uncertainty quantification capabilities of Gaussian processes.
1070 This work was originally published in [?].

1071 The framework implements Deep Gaussian Processes as a composition of Gaussian
1072 process layers, where each layer transforms its inputs through a Gaussian process
1073 mapping. This hierarchical structure allows the model to capture complex,
1074 non-stationary patterns in data while maintaining uncertainty estimates
1075 throughout the entire processing pipeline. GPflux handles this complexity through
1076 a carefully designed architecture that separates the concerns of model
1077 specification, inference, and prediction.

1078 A key innovation in GPflux is its integration with Keras-style layers, allowing
1079 practitioners to combine deterministic neural network layers with Gaussian
1080 process layers in a single model. This hybrid approach enables the construction of
1081 models that can leverage both the structural assumptions encoded in neural
1082 network architectures and the flexibility of Gaussian processes.

1083 For inference, GPflux implements various approximate methods, including
1084 stochastic variational inference with inducing points, which makes it possible to
1085 train deep Gaussian process models on large datasets. The framework provides
1086 built-in support for minibatch training, making it practical to work with datasets
1087 that don’t fit in memory.

3.3 Experimental Setup

Synthetic datasets were used to evaluate the proposed methods in classification and regression tasks, following very similar procedures to the ones used in [?], which are as follows:

3.3.1 Classification

A fully synthetic data for a 1-D ($P = 1$) multiclass classification problem with $K = 3$ classes is generated. The input feature matrix $\mathbf{X}$ is constructed by randomly sampling $N = 100$ points from a uniform distribution over the interval $[0, 1]$. For each sample n , the true label is determined by computing three target values:

$$\begin{aligned} t_{n,1} &= \sin(2\pi x_n), \\ t_{n,2} &= -\sin(2\pi x_n), \\ t_{n,3} &= -\sin(2\pi(x_n + 0.25)) + 0.5, \end{aligned} \tag{3-7}$$

and assigning the class label as the index $i \in \{1, 2, 3\}$ that maximizes $t_{n,i}$. This construction ensures a nontrivial, nonlinear decision boundary between the classes [?].

For evaluation, the test set is generated by extracting 200 equally spaced samples from the interval $[0, 1]$, providing a dense coverage of the input space to assess generalization performance.

Note that at this point, the fully synthetic dataset does not contain real annotator labels. Therefore, it is necessary to simulate annotator labels as corrupted versions of the hidden ground truth. In these experiments, the simulation is designed to account for: 1) dependencies among annotators, and 2) labeler performance that varies as a function of the input features.

To achieve this, an SLFM-based approach (termed SLFM-C) is employed, which generates annotator labels as follows:

- 1110 **1. Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q :$
 1111 $\mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R, n \in N$.
- 1112 **2. Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$,
 1113 where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D
 1114 representation of the input features in $\mathbf{X}$, obtained using the t-distributed
 1115 Stochastic Neighbor Embedding (t-SNE) approach.
- 1116 **3. Reliability calculation:** Calculate $\hat{\lambda}_n^r = \varsigma(\hat{f}_{l,r,n})$, where $\varsigma(\cdot) \in [0, 1]$ is the
 1117 inverse exponential sigmoid function.
- 1118 **4. Label assignment:** The r th annotator's label for the n th sample is given by

$$y_n^r = \begin{cases} y_n, & \text{if } \hat{\lambda}_n^r \geq 0.5 \\ \tilde{y}_n, & \text{if } \hat{\lambda}_n^r < 0.5 \end{cases}$$
 1119 where y_n is the true label and $\tilde{y}_n$ is a flipped version of the actual label y_n .

1120 This procedure ensures that the simulated annotator labels exhibit both
 1121 inter-annotator dependencies and input-dependent reliability.

1122 3.3.2 Regression

1123 A fully synthetic dataset is also generated for a 1-D regression problem, where the
 1124 ground truth for the n th sample is defined as:

$$y_n = \sin(2\pi x_n) \sin(6\pi x_n) \quad (3-8)$$

1125 Here, the input matrix $\mathbf{X}$ is constructed by randomly sampling 100 points uniformly
 1126 from the interval $[0, 1]$. The test set is created by extracting equally spaced samples
 1127 from the same interval, ensuring a clear separation between training and testing
 1128 data.

1129 To simulate the presence of multiple annotators, the ground truth is corrupted with
 1130 annotator-specific noise. For each annotator r , the observed label for the n th sample
 1131 is given by:

$$y_n^r = y_n + \epsilon_n^r$$

1132 where $\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$ is Gaussian noise, and v_n^r represents the error variance for
 1133 the r th annotator, which may depend on the input features. This setup allows us
 1134 to model both the bias and the reliability of each annotator, as well as potential
 1135 dependencies between annotators.

1136 To further increase realism, the error variance v_n^r can be modeled as a function of
 1137 the input features, capturing scenarios where annotator reliability varies across the
 1138 input space, thus:

$$\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$$

1139 being v_n^r , the r th annotator's error variance for the n th sample.

1140 Again, for modeling the error as a function of the input features, an **SLFM**-based
 1141 approach (termed SLFM-R) is employed to build the noisy labels, as follows:

- 1142 **1. Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q :$
 1143 $\mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R, n \in N$.
- 1144 **2. Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$,
 1145 where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D
 1146 representation of the input features in $\mathbf{X}$, obtained using the t-distributed
 1147 Stochastic Neighbor Embedding (t-SNE) approach.
- 1148 **3. Compute variance :** Finally, determine $v_n^r = \exp(\hat{f}_{l,r,n})$.

3.4 Results

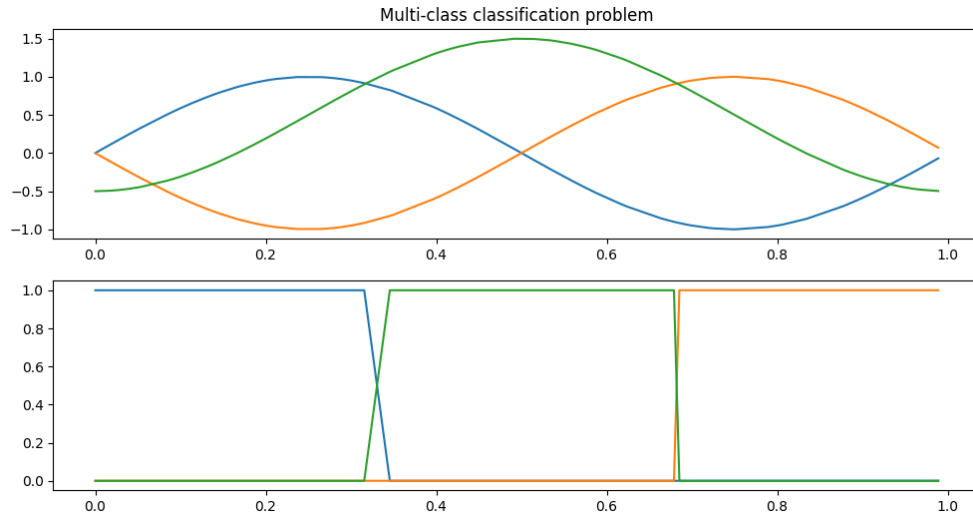
The built synthetic datasets are used to train the **CCGPMA** model. Emulated classification labels following the true label determining function from ?? can be seen in Figure ?. On the other hand, the ground truth and noisy labels emulation for annotators across the feature space for regression task can be seen in Figure ?.

Using both GPflux and GPflow for the regression and classifications tasks, leads to the following combinations experiments: GPflux-CCGPMA Regression, GPflow-CCGPMA Regression, GPflux-CCGPMA Binary Classification, GPflow-CCGPMA Binary Classification, GPflow-CCGPMA Multiclass Classification, GPflux-CCGPMA Multiclass Classification. All of the experiments converged successfully with almost identical results and close to no difference in performance.

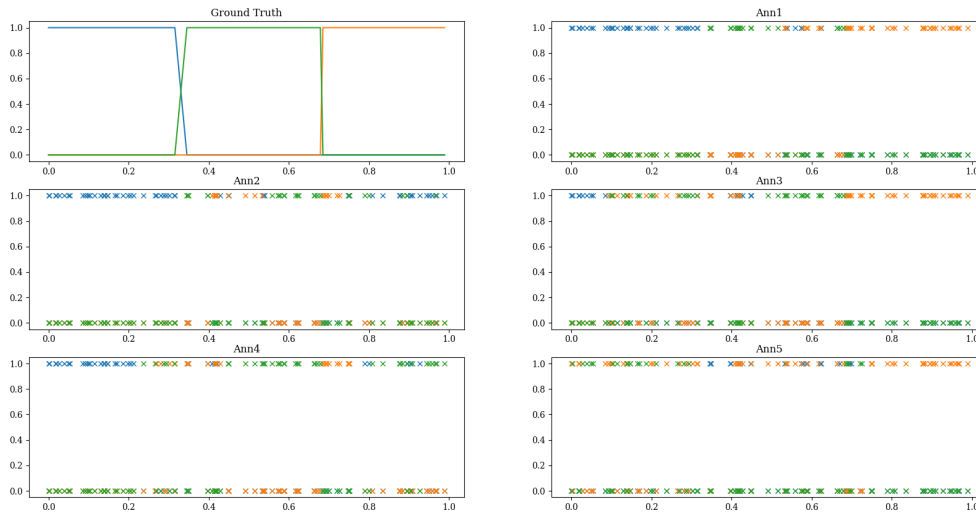
3.5 Conclusion

3.5.1 Comparative Analysis

The evolution of multiple annotator learning methods represents a fascinating progression in how we approach the challenge of learning from diverse expert opinions. **KAAR** established a foundation with its non-parametric approach through kernel alignment, though it operates under the assumption of constant annotator performance across the input space. **LKAAR** advanced this concept by introducing input-dependent annotator behavior, recognizing that expert performance often varies across different types of inputs. **RCDNN** brought the power and flexibility of deep learning to the problem, offering excellent scalability but requiring careful attention to regularization and architecture design. **CCGPMA** represents perhaps the most theoretically elegant approach, providing a principled



(a) Ground truth labels across the feature space



(b) Noisy emulations for annotators across the feature space for classification task

Figure 3-1 Synthetic dataset overview for classification task

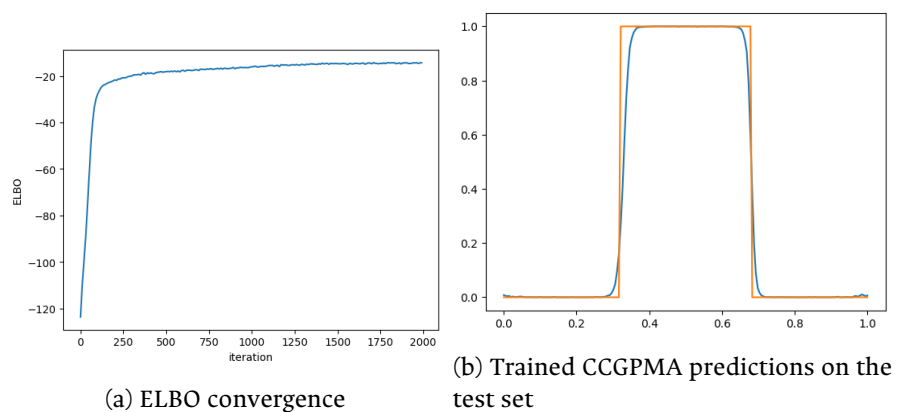


Figure 3-2 Classification results for the CCGPMA model using GPflux.

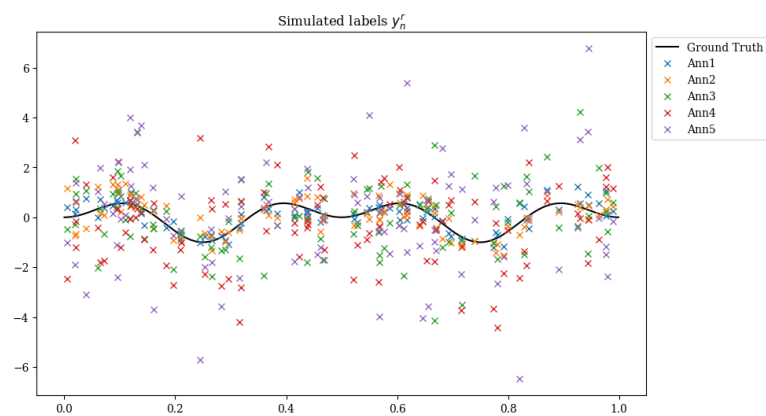
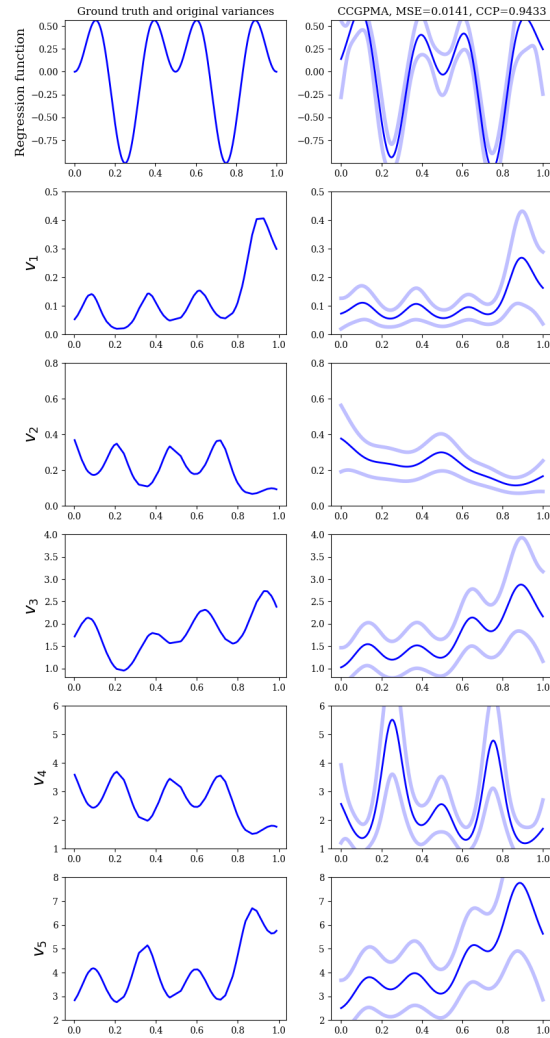


Figure 3-3 Ground truth and noisy labels emulation for annotators across the feature space for regression task.



(a) Trained CCGPMA predictions on the test set, for inference of both the hidden ground truth and the annotators' reliabilities across the feature space.

Figure 3-4 Regression results for the CCGPMA model using GPflux.

1173 probabilistic framework, though it faces significant computational challenges in
1174 practical applications.

1175 The selection of an appropriate method for a given application requires careful
1176 consideration of several factors. The size and dimensionality of the dataset play a
1177 crucial role, as some methods scale better than others with increasing data
1178 volume. The need for uncertainty quantification may favor probabilistic
1179 approaches like CGP, while computational constraints might push towards more
1180 efficient methods like KAAR. The presence of input-dependent annotator behavior
1181 could make LKAAR or RCDNN more appropriate choices. Additionally, when
1182 interpretability is a key requirement, as in many medical applications, methods
1183 like KAAR and LKAAR might be preferred over the more complex Deep Learning
1184 approaches. Table ?? provides a comprehensive comparison of the key
1185 characteristics and capabilities of each method discussed.

Table 3-1 Comparison of Multiple Annotator Learning Methods

Characteristic	KAAR [?]	LKAAR [?]	RCDNN [?]	CGP [?]
Input-dependent annotator behavior	No	Yes	Yes	Yes
Annotator dependencies modeling	Yes	Yes	Yes	Yes
Uncertainty quantification	No	No	Partial	Yes
Computational scalability	High	Medium	High	Low
Missing data handling	Yes	Yes	Yes	Yes
Non-linear relationships	Yes	Yes	Yes	Yes
Training complexity	Low	Medium	High	High
Hyperparameter sensitivity	Low	Medium	High	Medium
Interpretability	High	Medium	Low	Medium
Memory requirements	Low	Medium	High	High

Note: Partial uncertainty quantification in RCDNN refers to the use of Monte Carlo dropout for approximate uncertainty estimation.

1186 3.5.2 Implementation Considerations

1187 When implementing Chained Gaussian Processes for multiple annotator learning
1188 using these tools, several key considerations must be addressed. First, the model
1189 architecture must be carefully designed to capture both the individual annotator
1190 characteristics and their interdependencies [?]. This requires extending the basic
1191 GP layers provided by GPflux to handle multiple output streams - one for each
1192 annotator's reliability model.

1193 Second, the inference procedure must be adapted to handle the unique challenges
1194 of multiple annotator data, such as missing labels and varying levels of annotator
1195 expertise across different regions of the input space. The variational inference
1196 capabilities of GPflux provide a solid foundation for this, but careful attention
1197 must be paid to the design of the variational approximation to ensure it captures
1198 the relevant uncertainty in both the ground truth and annotator reliability
1199 estimates.

1200 Third, kernel selection and design presents a critical challenge in implementing the
1201 CGP model. The choice of kernel functions significantly impacts the model's ability
1202 to capture the underlying structure of the data and the relationships between
1203 annotators [?]. Several considerations must be taken into account:

- 1204 • **Kernel Complexity:** The model requires multiple kernel functions to capture
1205 different aspects of the data - one for the ground truth and additional kernels
1206 for each annotator's reliability model. This multi-kernel setup increases the
1207 complexity of kernel selection and optimization.
- 1208 • **Input-Dependent Behavior:** Since annotator performance often varies across
1209 the input space, the kernel functions must be flexible enough to capture this
1210 non-stationary behavior. This may require using composite kernels or kernels
1211 with input-dependent lengthscales.

- 1212 • **Inter-Annotator Dependencies:** The kernel design must account for potential
1213 correlations between different annotators' behaviors. This can be achieved
1214 through careful design of the kernel structure in the SLFM framework, where
1215 the combination of basis kernels must be chosen to effectively model these
1216 dependencies.
- 1217 • **Computational Efficiency:** The choice of kernel functions also impacts
1218 computational performance. While more expressive kernels may better
1219 capture complex patterns, they often come with increased computational
1220 cost. This trade-off must be carefully considered, especially when dealing
1221 with large datasets.

1222 Finally, computational efficiency becomes a critical concern when scaling to large
1223 datasets with multiple annotators. The minibatch training capabilities of both
1224 frameworks help address this, but additional considerations such as appropriate
1225 inducing point selection and batch size tuning become important for achieving
1226 good performance in practice.

1227 3.5.3 Final Remarks

1228 In this chapter, we have presented a comprehensive overview of the Chained
1229 Gaussian Processes model and its application to multiple annotator learning. We
1230 have discussed the theoretical underpinnings of the model, its implementation
1231 using GPflow and GPflux, and its evaluation on synthetic datasets. It was shown in
1232 [?] that **CCGPMA** outperforms other method in state of the art methods in terms of
1233 accuracy and uncertainty quantification (see ??).

1234 While Gaussian Processes have demonstrated remarkable success in regression
1235 and classification tasks, their application to image segmentation presents
1236 significant computational challenges [?]. The cubic computational complexity of
1237 GPs ($\mathcal{O}(N^3)$), which scales with the number of data points, becomes particularly

1238 problematic when dealing with high-resolution images where each pixel can be
1239 considered a data point [?]. This limitation makes traditional GP approaches
1240 impractical for large-scale image segmentation tasks, where the number of pixels
1241 can easily reach into the millions. Alternative approaches, such as deep
1242 learning-based segmentation methods, often provide more computationally
1243 efficient solutions for these high-dimensional problems, though at the cost of
1244 uncertainty quantification capabilities that GPs naturally provide [?] [?].

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CHAPTER

FOUR

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TRUNCATED GENERALIZED CROSS ENTROPY FOR SEGMENTATION

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4.1 Loss Functions for Multiple Annotators

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As mentioned in Section ??, a loss function is a key element for defining the objective function of a deep learning model. The categorical cross-entropy loss is a common loss function for classification tasks. However, in the case of multiple annotators, the categorical cross-entropy loss is not able to handle the varying reliability of the annotators. In this section, we will propose a loss function that is able to handle multiple annotators' segmentation masks while accounting for their varying reliability across different regions of the image.

1259 4.1.1 Generalized Cross Entropy

1260 The **Generalized Cross Entropy (GCE)** loss function was first introduced by [?] as a
 1261 robust alternative to the standard cross-entropy loss, particularly effective in
 1262 handling noisy labels. Let us first consider the **Cross Entropy (CE)** and **Mean**
 1263 **Absolute Error (MAE)** loss functions:

$$MAE(\mathbf{y}, f(\mathbf{x})) = \|\mathbf{y} - f(\mathbf{x})\|_1 \quad (4-1)$$

$$CE(\mathbf{y}, f(\mathbf{x})) = \sum_{k=1}^K y_k \log(f_k(\mathbf{x})) \quad (4-2)$$

1264 where $y_k \in \mathbf{y}$, $f_k(\mathbf{x}) \in f(\mathbf{x})$, and $\|\cdot\|_1$ stands for the l_1 -norm. Of note, $\mathbf{1}^\top \mathbf{y} =$
 1265 $\mathbf{1}^\top f(\mathbf{x}) = 1$, $\mathbf{1} \in \{1\}^K$ being an all-ones vector. In addition, the MAE loss can be
 1266 rewritten for softmax outputs, yielding:

$$MAE(\mathbf{y}, f(\mathbf{x})) = 2(1 - \mathbf{y}^\top \odot f(\mathbf{x})) \quad (4-3)$$

1267 where $\odot$ stands for the element-wise product.

1268 The **CE** loss function exhibits several distinct characteristics that make it
 1269 particularly sensitive to noisy labels. It is unbounded from above and heavily
 1270 penalizes confident but wrong predictions, which can lead to instability in the
 1271 presence of label noise. In contrast, the **MAE** loss offers a more robust alternative
 1272 with its bounded nature and symmetric properties in softmax-based
 1273 representations. While the MAE loss is more resilient to noisy labels, it tends to
 1274 train more slowly due to its equal weighting of all mistakes regardless of
 1275 confidence.

1276 The GCE loss function, as defined by the authors in [?], provides a flexible framework
 1277 that bridges these two approaches:

$$GCE(\mathbf{y}, f(\mathbf{x})) = 2 \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q}, \quad (4-4)$$

1278 with $q \in (0, 1]$. Remarkably, the limiting case for $q \rightarrow 0$ in GCE is equivalent to the
 1279 CE expression, and when $q = 1$, it equals the MAE loss. In addition, the GCE holds
 1280 the following gradient with regard to θ :

$$\frac{\partial GCE(\mathbf{y}, f(\mathbf{x}; \theta)|k)}{\partial \theta} = -f_k(\mathbf{x}; \theta)^{q-1} \nabla_\theta f_k(\mathbf{x}; \theta). \quad (4-5)$$

1281 The GCE loss combines the best of both worlds, offering robustness to label noise
 1282 while maintaining the convexity property necessary for optimization. The
 1283 truncation parameter q provides a mechanism to control the sensitivity to outliers,
 1284 allowing for fine-tuning of the loss function's behavior based on the specific
 1285 characteristics of the dataset.

1286 4.1.2 Extension to Multiple Annotators

1287 In the context of multiple annotators, we need to consider the varying reliability
 1288 of each annotator across different regions of the image. Let's consider a k -class
 1289 multiple annotators segmentation problem with the following data representation:

$$\mathbf{X} \in \mathbb{R}^{W \times H}, \{\mathbf{Y}_r \in \{0, 1\}^{W \times H \times K}\}_{r=1}^R; \hat{\mathbf{Y}} \in [0, 1]^{W \times H \times K} = \mathbf{f}(\mathbf{X}) \quad (4-6)$$

1290 where the segmentation mask function maps the input to output as:

$$\mathbf{f}: \mathbb{R}^{W \times H} \rightarrow [0, 1]^{W \times H \times K} \quad (4-7)$$

1291 The segmentation masks $\mathbf{Y}_r$ satisfy the following condition for being a softmax-like
 1292 representation:

$$\mathbf{Y}_r[w, h, :] \mathbf{1}_k^\top = 1; \quad w \in W, h \in H \quad (4-8)$$

1293 4.1.3 Reliability Maps and Truncated GCE

1294 The key innovation in our approach is the introduction of reliability maps Λ_r for
1295 each annotator:

$$\left\{ \Lambda_r(\mathbf{X}; \theta) \in [0, 1]^{W \times H} \right\}_{r=1}^R \quad (4-9)$$

1296 These reliability maps serve as a mechanism to estimate the confidence of each
1297 annotator at every spatial location (w, h) in the image. By learning these maps
1298 jointly with the segmentation model, the network gains the ability to adapt to the
1299 varying levels of expertise across different regions of the image. This approach
1300 allows for dynamic weighting of each annotator's contribution based on their
1301 estimated reliability in specific areas, effectively handling cases where annotators
1302 might demonstrate different levels of expertise in different parts of the image.

1303 The proposed Truncated Generalized Cross Entropy for Semantic Segmentation
1304 (TGCE_{SS}) combines the robustness of GCE with the flexibility of reliability maps.
1305 Let us first define the key components:

$$\mathcal{L}_{GCE}(\mathbf{Y}_r, f(\mathbf{X}; \theta)) = \mathbb{E}_k \left\{ \mathbf{Y}_r \odot \left(\frac{\mathbf{1}_{W \times H \times K} - f(\mathbf{X}; \theta)^{\odot q}}{q} \right) \right\} \quad (4-10)$$

$$\mathcal{L}_{prior} = \frac{\mathbf{1}_{W \times H} - \left(\frac{1}{K} \mathbf{1}_{W \times H} \right)^{\odot q}}{q} \quad (4-11)$$

1306 where $\mathcal{L}_{GCE}$ represents the GCE loss component and $\mathcal{L}_{prior}$ represents the uniform
1307 prior component. The complete TGCE_{SS} loss function is then defined as:

$$\begin{aligned} TGCE_{SS}(\mathbf{Y}_r, f(\mathbf{X}; \theta) | \mathbf{r}(\mathbf{X}; \theta)) = \mathbb{E}_{r,w,h} \left\{ \right. \\ \Lambda_r(\mathbf{X}; \theta) \odot \mathcal{L}_{GCE} + \tilde{C}(\mathbf{1}_{W \times H} - \Lambda_r(\mathbf{X}; \theta)) \odot \mathcal{L}_{prior}; \\ \left. w \in W, h \in H; r \in R \right\} \end{aligned} \quad (4-12)$$

where $q \in (0, 1)$ controls the MAE or CE level in the same way as in the GCE loss function. $\tilde{C}$ controls is an additional hyperparameter which controls the tolerance to unreliable annotations. The loss function consists of two main components:

- The first term weighted by Λ_r represents the GCE loss for regions where the annotator is considered reliable
- The second term weighted by $(1 - \Lambda_r)$ and $\tilde{C}$ provides a uniform prior for regions where the annotator is considered unreliable

A summary of the loss function mechanism is shown in Figure ??.

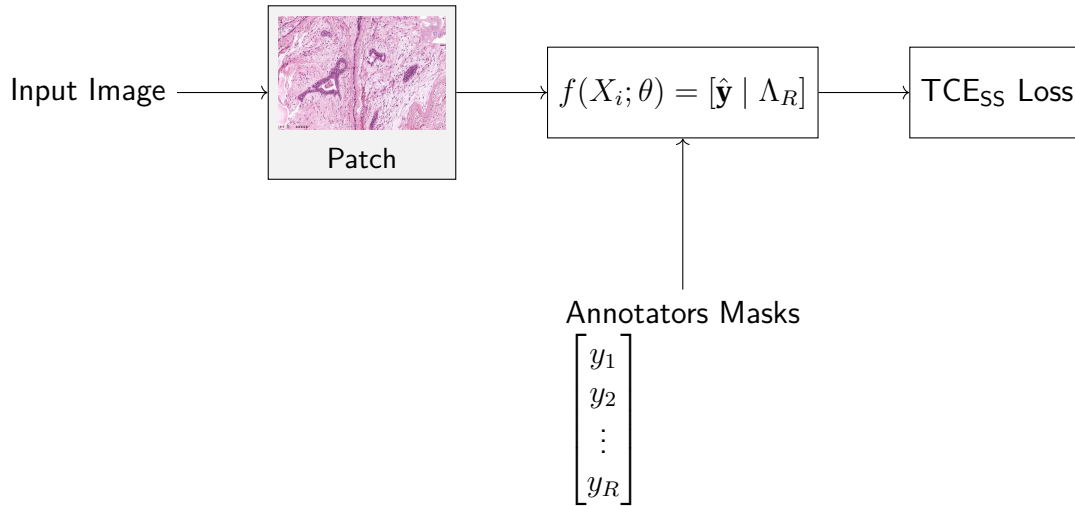


Figure 4-1 Working mechanism of the proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE_{SS}) loss function. The loss combines reliability maps Λ_r with the model's predictions $f(\mathbf{X}; \theta)$ to compute a weighted loss that accounts for annotator reliability across different image regions.

4.2 Chapter overview

This chapter presents a novel approach to handle multiple annotators in semantic segmentation tasks through the introduction of the Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE_{SS}) loss function. The key contributions and novelties of this work can be summarized as follows:

- 1321 • **Reliability-aware Learning:** Unlike traditional approaches that treat all
1322 annotators equally, the proposed method introduces reliability maps (Λ_r)
1323 that dynamically estimate the confidence of each annotator at every spatial
1324 location in the image. This allows the model to adapt to varying levels of
1325 expertise across different regions.
- 1326 • **Robust Loss Function:** The proposed $TGCE_{SS}$ combines the benefits of both
1327 Cross Entropy and Mean Absolute Error losses through the Generalized
1328 Cross Entropy framework. The truncation parameter q provides a flexible
1329 mechanism to control the sensitivity to noisy labels while maintaining the
1330 convexity property necessary for optimization.
- 1331 • **Unified Framework:** The approach provides a unified framework that
1332 handles both reliable and unreliable regions through a weighted
1333 combination of GCE loss and uniform prior. This is particularly important in
1334 medical imaging where annotator expertise can vary significantly across
1335 different anatomical structures.

1336 The main novelty of this approach lies in its ability to simultaneously learn the
1337 segmentation task while estimating annotator reliability in a spatially-aware
1338 manner. This represents a significant advancement over existing methods that
1339 either:

- 1340 • Treat all annotators equally without considering their varying expertise
- 1341 • Use global reliability scores that don't account for spatial variations in
1342 annotator performance
- 1343 • Rely on fixed, pre-defined reliability measures rather than learning them from
1344 data

1345 The proposed $TGCE_{SS}$ loss function represents a step forward in handling the
1346 complex nature of multiple annotator scenarios in semantic segmentation,
1347 particularly in medical imaging applications where expert knowledge can vary
1348 significantly across different regions of interest. This work provides a foundation
1349 for more sophisticated approaches to handling multiple annotators in deep
1350 learning-based segmentation tasks.

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CHAPTER

FIVE

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1356

CHAINED DEEP LEARNING AND $TGCE_{SS}$ LOSS FOR IMAGE SEGMENTATION

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As mentioned in Chapter ??, U-shaped Convolutional Neural Networks (CNNs) have been proven a good solution for Semantic segmentation (SS) tasks in medical images, due to their ability to capture both global and local information with small datasets.

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The proposed model architecture combines the strengths of U-Net with a ResNet-34 backbone, specifically designed to work with the $TGCE_{SS}$ loss function by providing a multi-output network which focuses on two tasks: **Multiclass Semantic Segmentation** and **Reliability Estimation**. The architecture is illustrated in Figure ??.

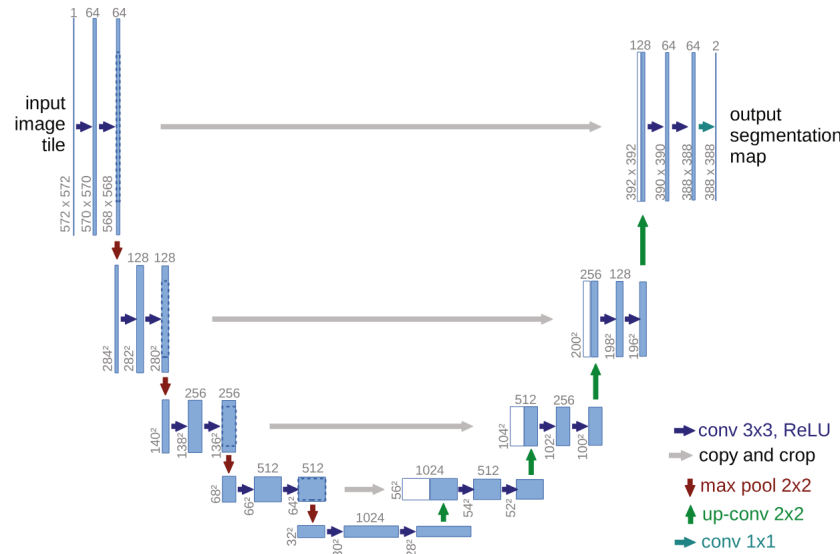


Figure 5-1 Original U-NET architecture from [?].

5.1 Backbone Architecture

The model employs a pre-trained ResNet-34 as its encoder backbone, leveraging its deep residual learning framework for efficient feature extraction. The choice of ResNet-34 provides several key advantages: efficient feature extraction through residual connections, pre-trained weights that capture rich visual representations, and stable gradient flow during training [?]. The ResNet-34 backbone is modified to serve as the encoder in our UNET architecture by removing the final fully connected layer and utilizing the feature maps from different stages of the network for skip connections.

5.1.1 U-Net Architecture

The U-Net architecture follows a traditional encoder-decoder structure with skip connections, where the encoder path implements the ResNet-34 structure. The

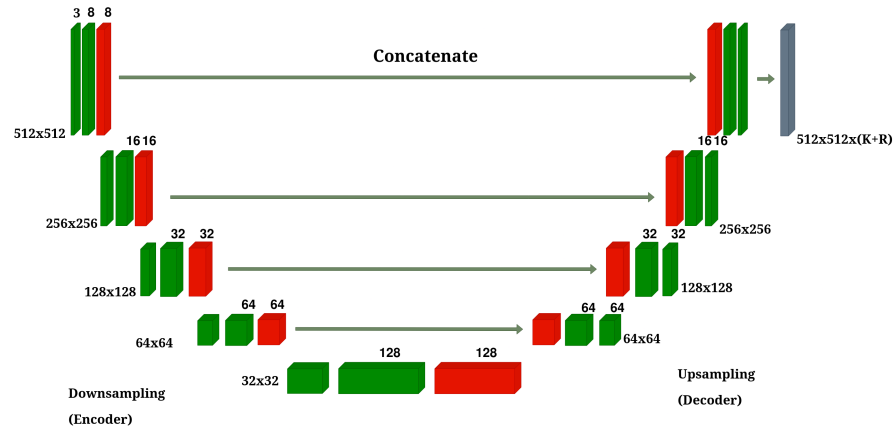


Figure 5-2 Overview of the proposed model architecture.

decoder path employs transposed convolutions for upsampling, creating a symmetrical architecture that effectively captures both high-level and low-level features. The architecture incorporates four downsampling stages in the encoder, corresponding to the ResNet-34 blocks, and four upsampling stages in the decoder. These stages are connected through skip connections that bridge corresponding encoder and decoder stages, allowing the network to preserve fine-grained details. Each convolution operation is followed by batch normalization and ReLU activation to ensure stable training and effective feature learning.

This U-Net architecture is expected to be coupled with a reliability branch which leads to a multi-output network. Three different approaches are considered for this branch: **scalar**, **features**, and **pixel**.

5.2 Reliability Branch

5.2.1 Scalar Reliability Estimation

The scalar reliability estimation branch is the simplest one, in which the reliability branch is a parallel branch to the segmentation branch, which leads to a

1393 multi-output network. The final output consists of R reliability maps Λ_r , one for
 1394 each annotator, with values constrained to the $[0, 1]$ range through a sigmoid
 1395 activation function. This design allows the network to learn and adapt to the
 1396 varying reliability of different annotators across different regions of the image.

1397 The scalar reliability branch is implemented by taking the bottleneck features from
 1398 the encoder and applying a global average pooling operation followed by a dense
 1399 layer with sigmoid activation. This results in a single reliability value per annotator
 1400 for each input image, making it computationally efficient but less spatially aware of
 1401 reliability variations within the image.

1402 5.2.2 Feature-based Reliability Estimation

1403 The feature-based reliability estimation approach provides a more detailed spatial
 1404 understanding of annotator reliability by operating on the bottleneck features of
 1405 the encoder. This approach maintains the spatial information from the encoder's
 1406 bottleneck while being more computationally efficient than pixel-wise reliability
 1407 estimation.

1408 The feature-based reliability branch is implemented by applying two convolutional
 1409 layers with batch normalization to the bottleneck features:

- 1410 • A first convolutional layer with 32 filters to process the bottleneck features
- 1411 • A second convolutional layer that produces R feature maps, one for each
- 1412 annotator

1413 This approach provides a coarse-grained spatial reliability map that captures
 1414 regional variations in annotator reliability while maintaining computational
 1415 efficiency. The resulting reliability maps have the same spatial dimensions as the
 1416 bottleneck features, providing a middle ground between scalar and pixel-wise
 1417 reliability estimation.

5.2.3 Pixel-wise Reliability Estimation

The pixel-wise reliability estimation approach provides the most detailed spatial understanding of annotator reliability by operating on the full-resolution features from the decoder. This approach allows for fine-grained reliability estimation at the pixel level, capturing local variations in annotator reliability.

The pixel-wise reliability branch is implemented by applying three convolutional layers with batch normalization to the decoder's output features:

- Two initial convolutional layers with 32 filters each to process the features
- A final convolutional layer that produces R reliability maps, one for each annotator

This approach provides the most detailed spatial reliability information, allowing the model to capture pixel-level variations in annotator reliability. The resulting reliability maps have the same spatial dimensions as the input image, providing the finest granularity of reliability estimation among the three approaches.

5.3 Integration with TGCE Loss

The model produces two distinct outputs: segmentation masks $\mathbf{\hat{Y}} = f(\mathbf{X}; \theta)$ and reliability maps $\{\Lambda_r(\mathbf{X}; \theta)\}_{r=1}^R$. These outputs work in tandem with the $TGCE_{SS}$ loss function described in Section ???. The loss function simultaneously guides the learning of both the segmentation masks and reliability maps, ensuring that the model learns to balance the contributions of different annotators based on their estimated reliability.

The $TGCE_{SS}$ loss function is implemented differently for each reliability approach, with specific adaptations to handle the different spatial dimensions of the reliability maps. Naturally, the model itself contains tunable hyperparameters related to the **Generalized Cross Entropy (GCE) - Mean Absolute Error (MAE)** trade off behavior (also known as q parameter) and the unreliability tolerance $\tilde{C}$, however, some additional hyperparameters were added with the purpose of regularizing the overall loss for every experiment.

1446 **5.3.1 Regularization terms**

1447 All three implementations include regularization terms to ensure proper learning of
1448 the reliability maps:

1449 **Numerical stability**

1450 To ensure numerical stability in the loss function computations, particularly in
1451 division operations, a small epsilon value ϵ is added to denominators. This
1452 prevents division by zero errors and stabilizes gradient calculations during
1453 training. The epsilon value is kept small enough to not significantly impact the loss
1454 function behavior while providing the necessary numerical safeguards.

1455 **Penalty for extreme reliabilities**

1456 Some optimization processes might favor a convergence to extreme reliability
1457 estimations, by assigning either 0 or 1 to all labelers. To avoid this, a regularization
1458 term is added to the loss function which penalizes the model for assigning extreme
1459 reliability values and favoring those around a centroid. High scaling values for this
1460 regularization term might lead to a convergence to a very homogeneous reliability
1461 map across all labelers, which is not desired either.

$$\mathcal{L}_{centroid} = \alpha_{centroid} \cdot \mathbb{E}[(\Lambda_r - 0.5)^2] \quad (5-1)$$

1462 **Entropy regularization**

1463 In this case, entropy is useful to ensure that the reliability maps are not too
1464 homogeneous. The entropy regularization term encourages the model to assign
1465 more discriminative reliability values by penalizing uniform distributions. This
1466 prevents the model from converging to uninformative reliability estimates where
1467 all labelers appear equally reliable, thus maintaining meaningful distinctions
1468 between more and less trustworthy annotators.

$$\mathcal{L}_{entropy} = \alpha_{entropy} \cdot \mathbb{E}[\Lambda_r \log(1 + \Lambda_r) + (1 - \Lambda_r) \log(1 + (1 - \Lambda_r))] \quad (5-2)$$

Sum-to-one regularization

The reliability branch for scalar, pixel and features approaches, might or might not be forced to sum to one through a softmax operation¹. For the cases in which this sum is not forced to be unitary the flexibility of this sum is manageable through a regularization term which penalizes the model for not summing to one. This was found useful in situations in which the reliability maps were not expected to be punctuated (e.g. for the scalar reliability) as $1/R$, but rather close to the unit for all of them (if all labelers were good). This regularization term was found particularly useful in situations where labelers have good and similar levels of expertise.

$$\mathcal{L}_{sum} = \alpha_{sum} \cdot \mathbb{E}[\sum_{r \in R} (\Lambda_r - 1)^2] \quad (5-3)$$

Missing labels penalizing

In situations where the datasets has a low labeling rate, the model might take an optimization shortcut by assigning high reliability values to labelers which did not label at all, since the element-wise product with the mask would be close to zero, which favors a minimization target, which is not desired. To avoid this, a term was added to the loss function which penalizes the model for assigning high reliability values to unlabeled patches, as follows:

$$\mathcal{L}_{\lambda} = \gamma \max \left(0, \sum_{\forall r \in R} \frac{1}{1 - \alpha_{\lambda}} (\Lambda_r - \mu_r) \cdot e^{\frac{\Lambda_r - \mu_r - 1}{\beta}} \right) \quad (5-4)$$

Where Λ_r is the estimated reliability of the labeler r for a certain patch, μ_r is the one hot label mask of the labeler r for the same patch, α_{λ} and β are hyperparameters which control the penalty strength for high and low estimated

¹This was left as flexible and configurable in the Network Architecture Building API from `seg_tgce`

1488 reliabilities, respectively and γ is a hyperparameter which controls the overall
 1489 strength of the penalty.

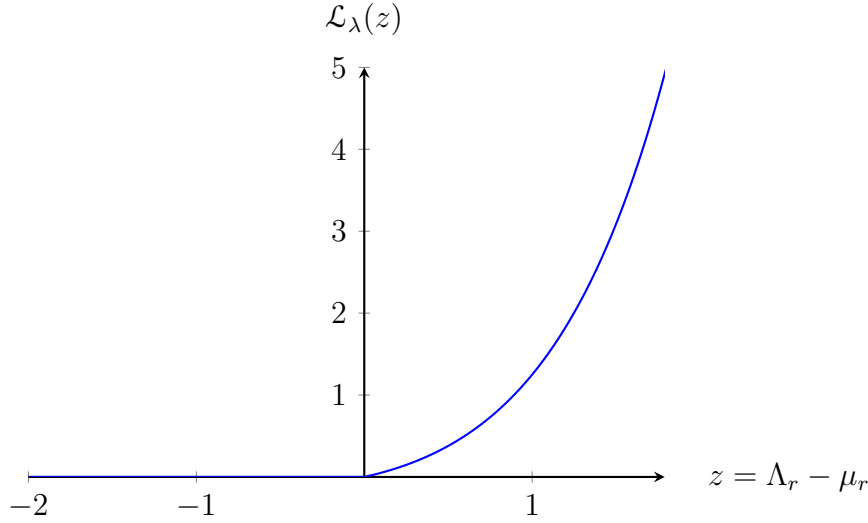


Figure 5-3 Example of the $\mathcal{L}_\lambda(z)$ function for values of $\alpha_\lambda = 0.2$ and $\beta = 1.0$.

1490 This function applies no penalty for cases in which $\Lambda_r < \mu_r$, meaning no penalty
 1491 for assigning any reliability value (ranging from 0 to 1) to a labeler who actually
 1492 labeled a given patch. On the other hand, a penalty is indeed applied in cases where
 1493 $\Lambda_r > \mu_r$, meaning a penalty for assigning high reliability values to a labeler which
 1494 did not label at all. The proportion on how high and how quickly (in the space of
 1495 difference $\Lambda_r - \mu_r$) the penalty is applied is controlled by the hyperparameters α_λ
 1496 and β . The behavior of this term is shown in Figure ??.

1497 These regularization terms lead to the overall loss function being in the form:

$$\begin{aligned}
L_{TGCE_{reg}} = L_{TGCE_{ss}} & \quad (5-5) \\
& + \alpha_{reg} \cdot \mathbb{E}[(\Lambda_r - 0.5)^2] \\
& + \alpha_{entropy} \cdot \mathbb{E}[\Lambda_r \log(1 + \Lambda_r) + (1 - \Lambda_r) \log(1 + (1 - \Lambda_r))] \\
& + \alpha_{sum} \cdot \mathbb{E}[\sum_{\forall r \in R} (\Lambda_r - 1)^2] \\
& + \gamma \cdot \max \left(0, \sum_{\forall r \in R} \frac{1}{1 - \alpha_\lambda} (\Lambda_r - \mu_r) \cdot e^{\frac{\Lambda_r - \mu_r - 1}{\beta}} \right)
\end{aligned}$$

1498 For the scalar reliability approach, the loss function operates on global reliability
 1499 values per annotator. The reliability maps are expanded to match the spatial
 1500 dimensions of the predictions through tiling operations. This approach is
 1501 computationally efficient but provides the same reliability value across all spatial
 1502 locations.

1503 The feature-based reliability approach requires additional processing to handle the
 1504 spatial dimensions of the bottleneck features. The reliability maps are resized
 1505 using bilinear interpolation to match the spatial dimensions of the predictions.
 1506 This approach provides coarse-grained spatial reliability information while
 1507 maintaining computational efficiency.

1508 The pixel-wise reliability approach operates directly on full-resolution reliability
 1509 maps, requiring no spatial transformations. This approach provides the most
 1510 detailed spatial reliability information but requires more computational resources.

1511

1512 5.4 Training Process

1513 The training process begins with the initialization of the ResNet-34 backbone
 1514 using pre-trained weights, providing a strong foundation for feature extraction.
 1515 The model's hyperparameters are optimized using Optuna [?], a hyperparameter
 1516 optimization framework that employs Bayesian optimization to search for optimal
 1517 values of:

- 1518 • Learning rate: Sampled logarithmically between 10^{-5} and 10^{-2}
- 1519 • TGCE_{SS} parameters:
 - 1520 – q : Controls truncation of the loss function (0.1 to 0.9)
 - 1521 – Noise tolerance: Controls model’s robustness to label noise (0.1 to 0.9)
 - 1522 – Regularization weights: Control the balance between different loss
 - 1523 terms.

1524 Two separate hyperparameter optimization studies were conducted using Optuna,
 1525 each with 20 iterations, on the Oxford PET dataset with three scorers emulating
 1526 SNR noises of -20, 0, and 10 dB respectively, and a labeling rate of 0.5. The first
 1527 study focused exclusively on regularization parameters with preselected values of
 1528 $q = 0.66$, $\tilde{C} = 0.62$, and learning rate = 6.5×10^{-4} , aiming to determine the
 1529 relative importance of different regularizers. The second study was
 1530 comprehensive, including the mandatory loss function parameters (q , $\tilde{C}$) and
 1531 learning rate, which are more sensitive to changes and directly affect the loss
 1532 function and optimizer behavior.

1533 Table ?? presents the hyperparameter importance ranking for the regularizers-only
 1534 study, while Table ?? shows the results from the comprehensive study including all
 1535 parameters.

Hyperparameter	Importance Score
β_λ	0.4728
α_{sum}	0.1364
α_{entropy}	0.1291
α_λ	0.1280
α_{centroid}	0.0997
γ_λ	0.0340

Table 5-1 Hyperparameter importance ranking for regularization parameters obtained from Optuna optimization study after 20 iterations. This study used preselected values of $q = 0.66$, $\tilde{C} = 0.62$, and learning rate = 6.5×10^{-4} . The importance scores indicate the relative impact of each regularization hyperparameter on the model’s performance.

Hyperparameter	Importance Score
learning_rate	0.2267
q	0.1832
$\tilde{C}$	0.1480
α_λ	0.1127
α_{sum}	0.0988
α_{entropy}	0.0868
α_{centroid}	0.0380
β_λ	0.0821
γ_λ	0.0237

Table 5-2 Comprehensive hyperparameter importance ranking including all parameters obtained from Optuna optimization study after 20 iterations. This study shows the relative importance of mandatory loss function parameters (q , $\tilde{C}$), learning rate, and regularization parameters together.

1536 After finding the optimal hyperparameters, the model is trained end-to-end using
 1537 the Adam optimizer with the best learning rate found ². The TGCE_{SS} loss function
 1538 plays a crucial role in updating both the segmentation and reliability branches,
 1539 ensuring that the model learns to effectively handle multiple annotators' inputs
 1540 while accounting for their varying reliability.

1541 The model's architecture is specifically designed to address the challenges of
 1542 multi-annotator segmentation. Through the ResNet-34 backbone, it learns robust
 1543 segmentation features that capture high-level patterns in the data. The UNET's
 1544 skip connections enable the preservation of fine-grained details, while the parallel
 1545 reliability branch allows the model to adapt to annotator-specific characteristics.
 1546 This comprehensive design enables the model to effectively handle multiple
 1547 annotators' inputs while maintaining high segmentation accuracy and reliability
 1548 estimation.

²For every experiment this was also tuned with the help of the Optuna study, having different optimal values for every experiment.

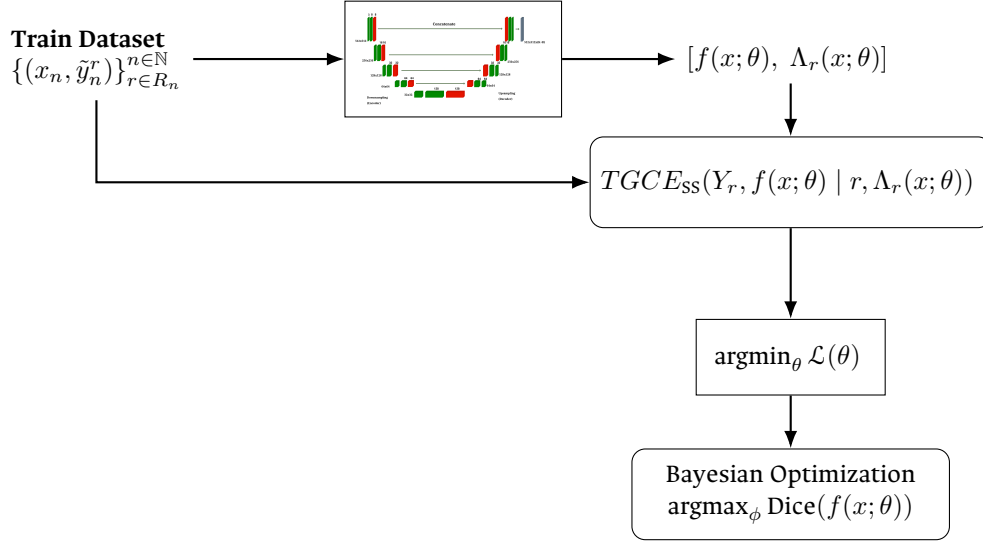


Figure 5-4 Training process of the proposed model overview. The model first undergoes hyperparameter optimization using Bayesian optimization, followed by end-to-end training with the optimal parameters. Estimated segmentation and reliability maps are computed for each input image, and the loss function is computed for the entire batch. Network parameters are tuned based on the loss function as objective (with minimization target) whereas the hyperparameters ϕ tuning are based on the Dice coefficient of the validation data set (with maximization target).

5.5 Experimental Setup

5.5.1 Noisy Annotations Generation

A challenge arises for the creation of emulated noisy annotations datasets, since it is expected for images annotations to have some degree of expertise variability, similar to what is expected in real crowdsourced datasets. Simply introducing random noise into the annotated masks does not work, since the original morphological structures from the expected ground truth are far from being preserved. Instead, a “noisy” labeler is expected to produce an annotation which has at least some degree of morphological consistency on itself, even if it shows discrepancies when compared with some metric (like DICE score) against the ground truth annotation.

1560 This has been proven experimentally when introducing random noise into any
 1561 popular segmentation dataset, in which the resulting mask is just a non coherent
 1562 map of noise across the image, as shown in Figure ??.

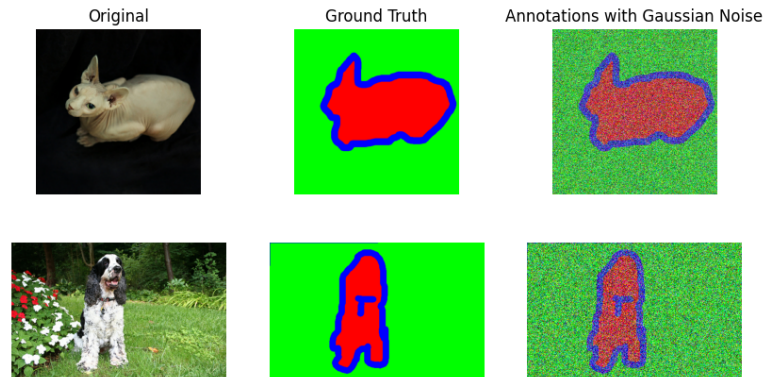


Figure 5-5 Example of a noisy mask generated by naively introducing random noise into a ground truth mask. Morphological consistency is lost.

1563 For this reason, a more sophisticated approach was needed to create datasets with
 1564 emulated noisy annotations. The approach used here was to use a pre-trained
 1565 U-Net model to produce a good enough ³ segmentation of the image, which was
 1566 then used to produce a “noisy” annotation by introducing random noise into the
 1567 last encoder layers of the model, thus preserving the morphological consistency of
 1568 the original annotation. This strategy works since slight modifications introduced
 1569 into the encoder layers weights, resemble conceptual disturbances with respect to
 1570 the original ground truth label, which kind of emulates the cultural behavior of
 1571 having a different interpretation of the image, either for having a different level of
 1572 expertise or for having a different point of view or school of thought. The first
 1573 encoding layers were not modified, since it is expected human labelers would
 1574 agree on the most fundamental structures (analog to extracted features from initial
 1575 convolutional layers) in a similar way, thus preserving some degree of
 1576 morphological consistency of the original annotation.

1577 In this way, the level of “disturbance” with respect to the ground truth for an

³More precisely, a U-Net architecture with Resnet-34 backbone model with a simple categorical cross-entropy loss which reached 0.8426 Dice Coefficient after 44 epochs.

emulated annotator can be controlled by the level of noise introduced into the weights of the last encoder layers, thus:

$$\mathbf{W}_{noisy} = \mathbf{W}_{original} + \epsilon_d \quad (5-6)$$

where $\mathbf{W}_{original}$ represents the original weights of the last encoder layers, ϵ_d follows a Gaussian distribution with zero mean and variance σ^2 . The variance σ^2 controls the level of noise introduced, and thus the degree of disturbance in the resulting segmentation masks.

With the application of the encoder layer weight perturbation technique, the resulting noisy masks are shown in Figure ?? . It can be seen that the morphological consistency is preserved, even though the resulting masks are far from the ground truth annotations, which goes perfectly well for testing the robustness of the models against noisy annotations in crowdsourcing-like scenarios.

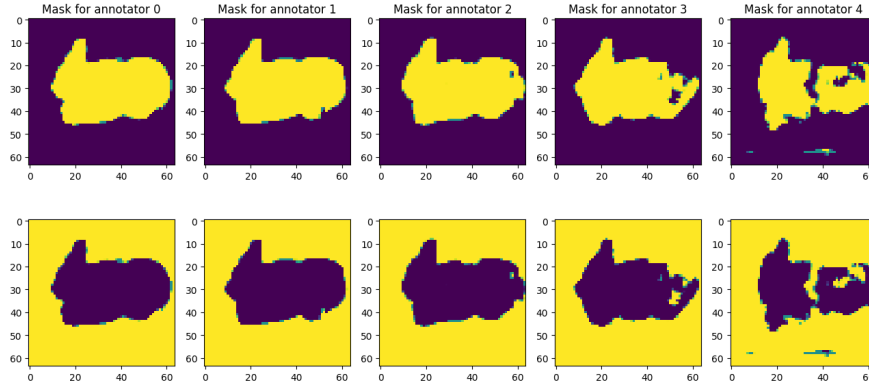


Figure 5-6 Noisy mask generated by enhancing the disturbances in the encoder layers weights for the Oxford-IIIT Pet Dataset. Morphological consistency is preserved. From left to right, SNR levels of noise in the encoder layer are 10, 5, 2, 0, -5 dB. Not adding noisy to the encoder layer would reach to an annotator behavior as good as the originally trained base model.

5.5.2 Missing Labels Emulation

It has also been determined that most real world crowdsourced datasets suffer from missing labels, since not all annotators are able to label all of the input samples. To emulate this behaviour, a missing labels dataset was created by randomly skipping labels from the annotators labels based on a labeling rate $r_l \in [0, 1]$. The labeling rate r_l controls the proportion of labels that are skipped from the annotators labels. The lower the labeling rate, the more labels are skipped from the annotators labels. The implementation of this missing labels emulation together with the noisy annotators generation can be flexibilized through an algorithm, which enables it to be used in experiments with variable amount of labelers, noise levels, and labeling rates. An overview of the algorithm is shown in Algorithm ???. A sample of the generated noisy and missing labels dataset is shown in Figure ??.

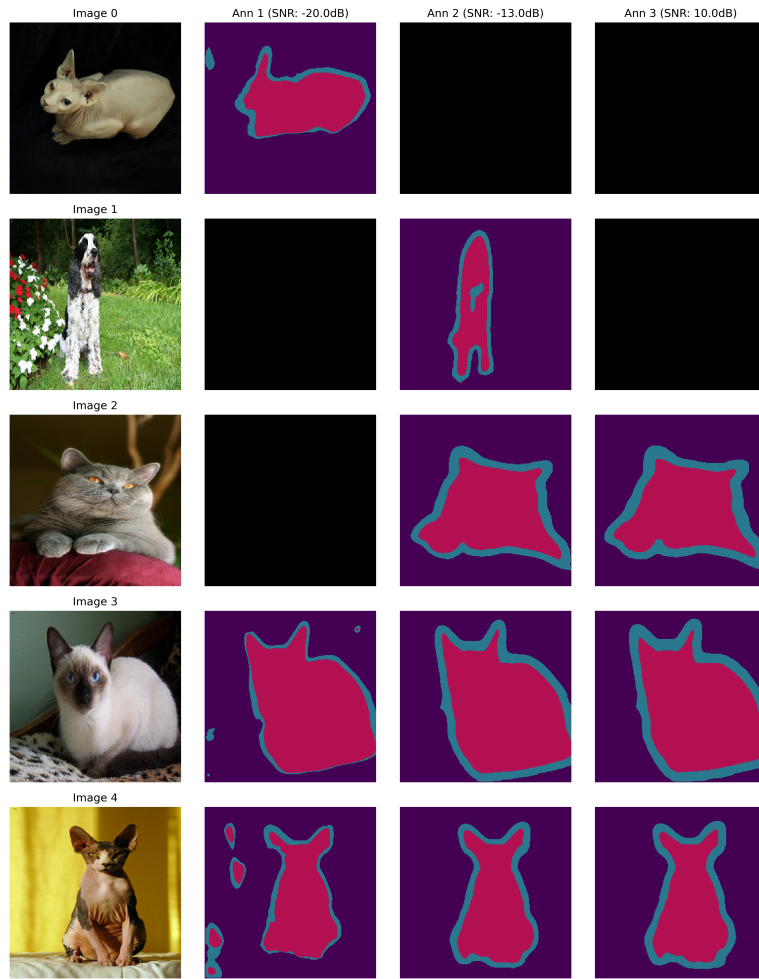


Figure 5-7 Example of missing labels and noisy annotations emulation for the Oxford-IIIT Pet Dataset. The same label rate ($r_l = 0.7$) is preserved for all labelers, which means in this case that every labeler is expected to label 70% of the samples, and skip the remaining 30%.

1601 5.6 Results

1602 Initially, a baseline model (ground truth only) was trained with an early stopping
 1603 mechanism, which stopped the training when the validation loss stopped decreasing
 1604 for 10 epochs, reaching only 20 epochs of significant training. The results are shown
 1605 in Figure ??.

Algorithm 5-1 Multi-Annotator Dataset Generation with Noisy and Missing Labels

Require: Base segmentation model M , SNR levels $\{s_1, \dots, s_n\}$, labeling rate $r \in [0, 1]$

Ensure: Dataset with multiple noisy annotations and missing labels

```

1: Step 1: Generate Noisy Annotators
2: for each SNR level  $s_i$  do
3:   Create disturbed model  $M_i$  by adding noise to  $M$  with SNR  $s_i$ 
4: end for
5: Step 2: Initialize Labeler Assignment System
6: Create assignment matrix  $A \in \{0, 1\}^{N \times n}$  where  $N$  is number of samples
7: for each annotator  $j \in \{1, \dots, n\}$  do
8:   for each sample  $i \in \{1, \dots, N\}$  do
9:      $A_{i,j} \leftarrow \begin{cases} 1 & \text{with probability } r \\ 0 & \text{with probability } 1 - r \end{cases}$ 
10:   end for
11: end for
12: Step 3: Generate Multi-Annotator Dataset
13: for each image  $x$  in dataset do
14:   Resize  $x$  to target dimensions
15:   for each annotator model  $M_i$  do
16:     Generate annotation  $y_i = M_i(x)$ 
17:     Apply labeler assignment mask:  $y_i \leftarrow y_i \cdot A_{i,j}$ 
18:   end for
19:   Store  $(x, \{y_1, \dots, y_n\})$  in dataset
20: end for
   return Multi-annotator dataset with noisy and missing labels

```

Then, using the scalar model described in Section ??, training was performed for 20 epochs and Bayesian hyperparameter optimization on learning rate, decoder dropout rate, for the regularizers described in Section ?. The results are shown in Figure ??.

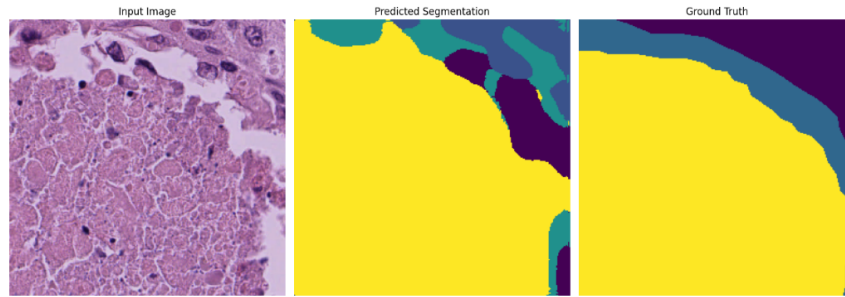


Figure 5-8 Segmentation results for baseline model on the Triple-Negative Breast Cancer (TNBC) (Crowd-Seg) dataset after 20 epochs of training. Even though the segmentation algorithm seems to show progress on its way to capture the underlying target structures to be segmented, since the ground truth annotation is not expressive enough, the Dice coefficient remains below 0.55, which is not considered a good performance. The baseline model remained with overfitting even after combining techniques of: Bayesian hyperparameter optimization, data augmentation, learning rate scheduling, dataset shuffling, and early stopping.

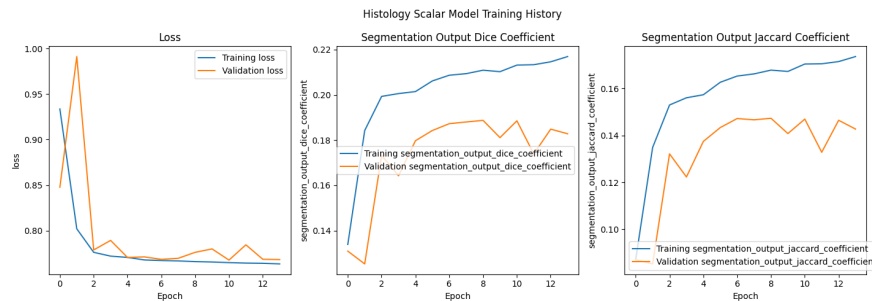


Figure 5-9 Segmentation results for scalar model on the TNBC (Crowd-Seg) dataset after 20 epochs of training. Again, training metrics show a constant grow, but not validation metrics, which shows a combined issue of overfitting and unreliable ground truth.

5.6.1 Oxford-IIIT Pet Dataset

The experimental results on the Oxford-IIIT Pet Dataset demonstrate interesting insights into the performance of different approaches. While STAPLE achieves the highest Dice and Jaccard scores (0.8718 and 0.7917 respectively) for the dataset with no missing labels (label rate of 1.0), our proposed TGCE_{SS} approaches show superior performance for label rates below 1.0 (Table ??). The pixel-wise variant, while slightly lower in performance metrics, provides more detailed spatial reliability information. It is of note that the key strength of TGCE_{SS} lies in its ability to simultaneously learn both segmentation and reliability estimation, providing valuable insights into annotator behavior and confidence levels that STAPLE and MV cannot offer.

Method	Label Rate 1.0		Label Rate 0.7		Label Rate 0.3	
	Dice	Jaccard	Dice	Jaccard	Dice	Jaccard
TGCE scalar	0.8622±0.009	0.7791±0.012	0.8568±0.017	0.7726±0.024	0.8529±0.004	0.7677±0.007
TGCE feature	0.6995±0.044	0.5733±0.047	0.4766±0.026	0.3441±0.027	0.4766±0.026	0.3441±0.027
TGCE pixel	0.8468±0.018	0.7589±0.025	0.8545±0.013	0.7695±0.017	0.8283±0.001	0.7359±0.002
MV	0.8132±0	0.7222±0	0.7572±0	0.6554±0	0.5566±0	0.4687±0
STAPLE	0.8718±0	0.7917±0	0.8439±0	0.7609±0	0.5734±0	0.5158±0

Table 5-3 Performance metrics (Dice and Jaccard) at different labeling rates. Best values per category are bolded.

The reliability estimation capabilities of our model are particularly noteworthy, as demonstrated in Figures ?? and ?. The model successfully captures the varying reliability levels of different annotators, with higher reliability assigned to annotators with lower noise levels (10 dB) and lower reliability to those with higher noise levels (-20 dB). This ability to quantify and visualize annotator reliability is a significant advantage over traditional approaches like STAPLE and MV, which focus solely on segmentation accuracy without providing insights into annotator behavior.

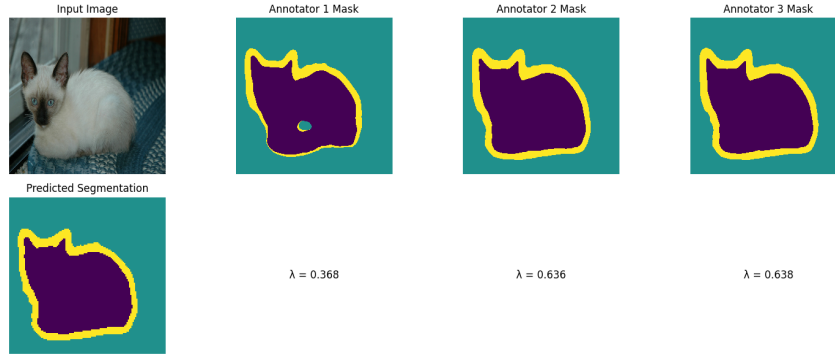


Figure 5-10 Example evaluation of the proposed model in its last trained epoch. An accurate segmentation output is delivered as well as scalar reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0011, $q = 0.4$, noise tolerance = 0.2, $\alpha_{reg} = 0.26$, $\alpha_{entropy} = 0.03$, $\alpha_{sum} = 0.27$.

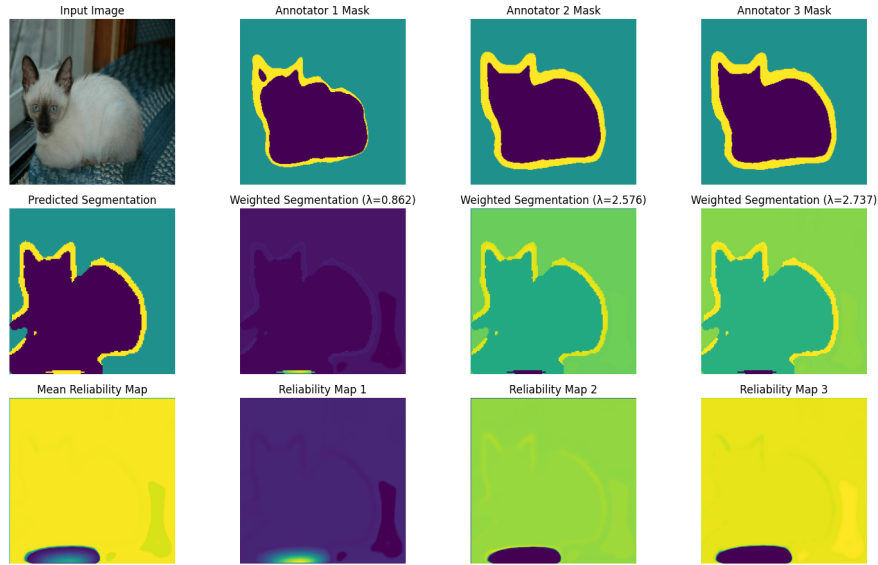


Figure 5-11 Example evaluation of the proposed model in some given epoch (5/50) of training, which demonstrates a quick convergence of the model for correctly estimating the reliability of the labelers masks. An accurate segmentation output is delivered as well as pixel-wise reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0027, $q = 0.9$, noise tolerance = 0.2, $\alpha_{reg} = 0.38$, $\alpha_{entropy} = 0.02$, $\alpha_{sum} = 0.42$.

5.6.2 TNBC Dataset

As mentioned in Section ??, [?] presented a **TNBC** dataset, which is crowdsourced and available as HistomicsTK metadata. Later, [?] sampled 151 **Whole Slide Imagings (WSIs)**, from which 161 **Region of Interests (ROIs)** were extracted and labeled by 20 medical students. The dataset was sampled as patches of size 512x512, resulting in 10173 training patches, 1264 validation patches and 399 testing patches. The segmentation masks for the patches are composed of the classes: Tumor, Stroma, Inflammation, Necrosis, and Others ⁴. This subset of the **TNBC** is named Crowd-Seg [?].

Raw PNG files for the images patches and masks were provided by the authors, in practice, in order to optimize the use of these data, a custom tensorflow dataset was built for facilitating batch memory, caching and prefetching. The custom dataset implementation handles the loading and preprocessing of both images and masks, including normalization of images to the [0,1] range and one-hot encoding of the segmentation masks. It also manages the multi-annotator structure by creating a labeler mask that indicates which annotators provided labels for each sample. The dataset supports data augmentation through random transformations (flips, rotations, brightness/contrast adjustments) and includes visualization utilities for debugging and analysis purposes. This custom dataset implementation is publicly available ^{5 6} within the PyPI package included in the overall solution of this work. (see Section ??).

Notes on the Crowd-Seg Dataset

The multiple labelers version of the **TNBC** dataset (Crowd-Seg) contains a couple of peculiarities that are worth mentioning. First, the label rate (inverse percentage of

⁴In practice, some of this masks also contain an additional class labeled as 'ignore'.

⁵https://github.com/blotero/seg_tgce

⁶<https://seg-tgce.readthedocs.io/en/latest/>

labels that are missing) is not uniform across the labelers and, in general, quite low, ranging from 1 to 12% as seen in Figure ??.

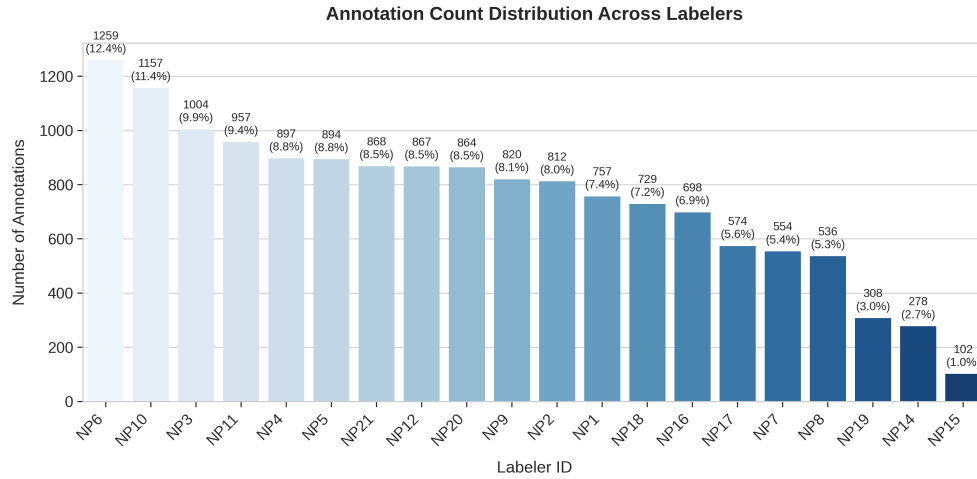


Figure 5-12 Label rate distribution across the labelers for the TNBC (Crowd-Seg) dataset.

On the other hand, the classes distribution across all patches are not equal, being ‘Tumor’ (class 2) the most frequent class. Despite the classes not being equally distributed, the distribution of the classes across the labelers is fairly uniform, as shown in Figure ??.

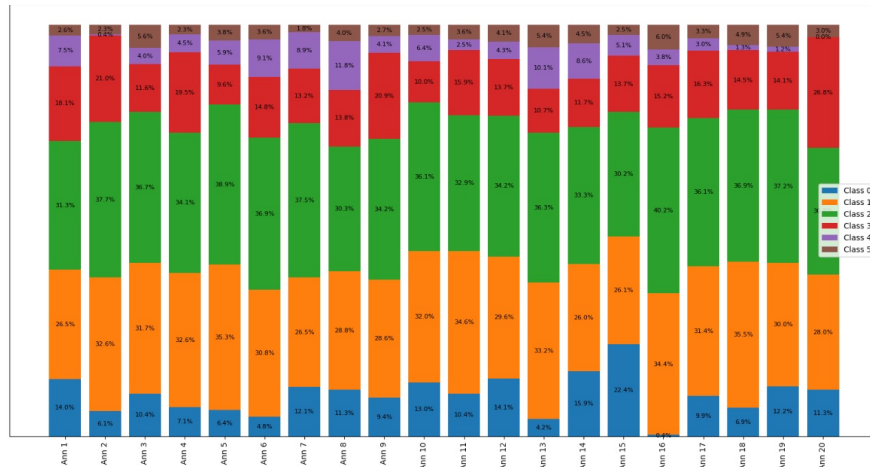


Figure 5-13 Classes distribution across the labelers for the TNBC (Crowd-Seg) dataset. Most common classes are ‘Tumor’ (class 2), ‘Other’ (class 1) and ‘Stroma’ (class 3).

1659 It was also found that the dice score of the labelers masks against the ground truth
 1660 differed across the labelers and the classes, as shown in Figure ???. The target class
 1661 (Tumor) ranged dice scores across the labelers from 0.5 to 0.9, other classes can
 1662 get as low as 0.4 for in average for some labelers, meaning very high levels of
 1663 disagreement.

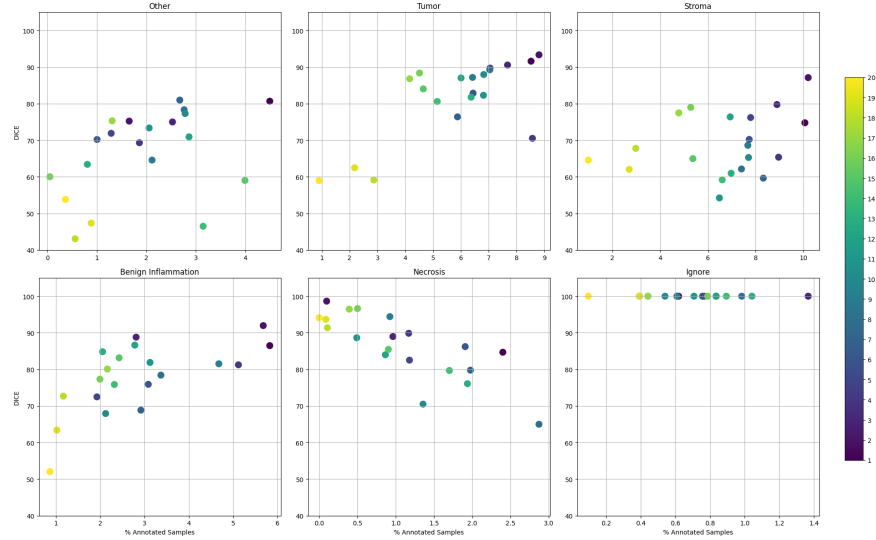


Figure 5-14 Dice score of the labelers masks against the ground truth for the TNBC (Crowd-Seg) dataset.

1664 Finally, it was found that the expert labels, which would thereby be considered as
 1665 ground truth for metrics evaluation, were not always correct, as shown in Figure
 1666 ??. These morphological inconsistencies make an objective metric for evaluating the
 1667 performance of the proposed model difficult to define, since the ground truth is not
 1668 always correct or expressive enough.

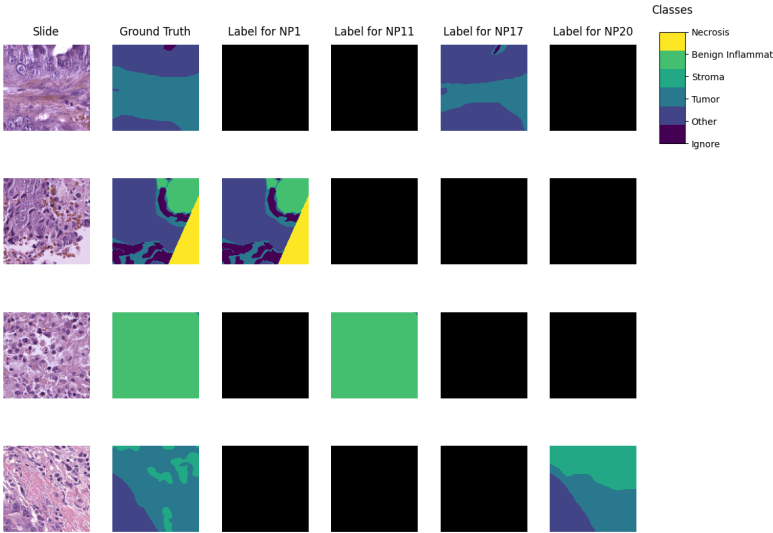


Figure 5-15 Example of ground truth with odd geometries (second patch from top to bottom), which are not always correct or compatible with the underlying structures of the patch itself.



Figure 5-16 Training curves for the baseline model on the TNBC (Crowd-Seg) dataset.

5.7 Summary

The experimental results on both the Oxford-IIIT Pet and TNBC (Crowd-Seg) datasets provide valuable insights into the performance and limitations of our

proposed approach. The model demonstrated exceptional performance on the Oxford-IIIT Pet dataset, achieving competitive Dice and Jaccard scores while successfully capturing annotator reliability patterns. The model's ability to simultaneously learn segmentation and reliability estimation proved particularly valuable, providing insights into annotator behavior that traditional approaches like STAPLE and MV cannot offer.

However, the experiments on the **TNBC** dataset revealed significant challenges in evaluating semantic segmentation models for histopathological images when dealing with unreliable ground truth. The morphological inconsistencies in the expert annotations, combined with the extremely low label rates (1-12%) across annotators, made it difficult to objectively measure the model's performance. While the model showed promising convergence in training, the validation metrics remained unstable, suggesting that the current dataset may not be suitable for rigorous evaluation of multi-annotator segmentation approaches.

These findings highlight the need for more reliable ground truth annotations in histopathological image segmentation datasets. Future work should focus on:

- Developing datasets with more consistent and reliable expert annotations
- Implementing stricter quality control measures for ground truth generation
- Exploring alternative evaluation metrics that account for annotation uncertainty
- Investigating methods to handle extremely sparse annotations more effectively

Despite these challenges, the model's performance on the Oxford-IIIT Pet dataset demonstrates its effectiveness in handling multiple annotators with varying reliability levels. The model successfully learned to:

- Balance the contributions of different annotators based on their reliability

- 1698 • Provide detailed spatial reliability information through pixel-wise estimation
- 1699 • Maintain high segmentation accuracy even with missing labels
- 1700 • Adapt to different noise levels in annotator inputs

1701 These capabilities make the proposed approach particularly valuable for
1702 real-world applications where multiple annotators with varying expertise levels
1703 contribute to the dataset. The model's ability to simultaneously learn
1704 segmentation and reliability estimation provides a more comprehensive
1705 understanding of the annotation process, which could be leveraged to improve the
1706 quality of future annotations and dataset construction.

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CHAPTER

SIX

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FINAL REMARKS

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6.1 Conclusions

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This thesis has presented a novel approach to address the challenges of Image Semantic Segmentation (ISS) in histopathology images, particularly in crowdsourcing scenarios where multiple annotators with varying levels of expertise contribute to the labeling process. The proposed solution successfully tackles the fundamental challenges of modeling annotator performance and handling label inconsistencies while maintaining computational efficiency. The work demonstrates significant improvements over existing state-of-the-art methods, particularly in scenarios with noisy and missing labels, which are common in crowdsourcing approaches.

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Throughout this work, several techniques have been explored to develop a model that can face Image Semantic Segmentation (ISS) of real world histopathology images in crowdsourcing scenarios. Different approaches were studied and evaluated to reach an optimal solution which could model the annotators' performance from the input space. The following remarks are based on the results obtained from the experiments conducted and reviewed literature:

- 1728 • Chained Gaussian Processes approaches, while robust for classification and
1729 regression tasks, proved computationally intensive for image segmentation.
1730 This limitation highlighted the need for more efficient approaches that could
1731 handle the complexity of medical image segmentation while modeling
1732 annotator performance.
- 1733 • The development of a novel loss function was crucial for the success of this
1734 work, as existing loss functions were insufficient to model the annotators'
1735 performance across the input space. The proposed loss function successfully
1736 enabled the model to adaptively infer the best possible segmentation without
1737 requiring separate inputs about labeler performance.
- 1738 • The introduction of a tensor map to codify labeler reliability proved to be an
1739 effective approach, allowing the model to implicitly weigh the labelers based
1740 on their performance across the mask and classes space, thus handling the
1741 labeler performance inconsistency effectively.
- 1742 • The use of U-Shaped Deep Learning models as a building block for the
1743 proposed solution was validated, as it effectively captured spatial
1744 relationships between pixels and annotations while handling the variability
1745 in image quality.
- 1746 • The combination of Chained Deep Learning approaches with the proposed
1747 loss function and tensor map representation resulted in a comprehensive
1748 solution being, to the best of our knowledge, the first approach on offering
1749 an image segmentation solution for multiple labelers in medical image with
1750 interpretability in the reliability of the labelers, successfully addressing both
1751 labeler inconsistency and image quality variability. The model outperformed
1752 other state of the art solutions in terms of segmentation accuracy in noisy
1753 and missing labels conditions, which is rather common in crowdsourcing
1754 approaches.

6.2 Future work

As tools and frameworks for bayesian approaches like Chained Gaussian Processes¹ evolve, it is expected to see a decrease in computational requirements and hence, making mixed models more appealing for the task of ISS of histopathology images. In that sense, the use of *Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA)* as a building block for the proposed solution could be a good approach, as it is a powerful model that can take advantage of the inter-dependencies between the annotators, and the powerfulness of CNNs to extract features from the images.

The noisy labelers emulation technique developed in this work presents an opportunity to test the model's robustness in various medical image segmentation challenges. This technique could be adapted to simulate different types of expert variability in other medical imaging domains:

- **Multi-modal Medical Imaging:** The technique could be extended to simulate expert disagreements in multi-modal scenarios, where different imaging modalities (e.g., MRI, CT, PET) might lead to varying interpretations of the same anatomical structure.
- **Temporal Consistency:** For longitudinal studies, the emulation could be modified to simulate how expert annotations might vary over time, helping to test the model's ability to maintain consistency in segmentation across different time points.
- **Cross-institutional Variability:** The noise injection technique could be calibrated to simulate the variations in annotation patterns that arise from different institutional protocols or training backgrounds.

¹Including GPFlow, and GPFlux.

1779 • **Pathology-specific Challenges:** The emulation could be tailored to simulate
1780 expert disagreements in specific pathological conditions, where the
1781 morphological features might be more ambiguous or subject to
1782 interpretation.

1783 These extensions would help validate the model's performance across a broader
1784 range of medical imaging scenarios and ensure its robustness in real-world clinical
1785 applications.

1786 The pixel-wise reliability mapping approach shows particular promise for future
1787 development. While the current implementation already demonstrates good
1788 performance, several enhancements could lead to significant improvements:

1789 • **Advanced Data Augmentation:** Implementing a comprehensive data
1790 augmentation strategy could significantly improve the model's convergence
1791 and performance. This could include:

- 1792 – Geometric transformations (rotations, flips, elastic deformations)
- 1793 – Intensity variations to simulate different staining conditions
- 1794 – Multi-scale training to handle varying tissue sizes
- 1795 – Adversarial training to improve robustness

1796 These augmentations would help the model learn more robust features and
1797 better generalize to unseen data.

1798 • **Attention Mechanisms:** Integrating attention mechanisms into the pixel-wise
1799 reliability branch could help the model focus on the most relevant regions for
1800 reliability estimation, potentially improving both accuracy and computational
1801 efficiency.

1802 • **Multi-scale Reliability Estimation:** Implementing a hierarchical approach to
1803 reliability estimation could capture both fine-grained local variations and
1804 broader regional patterns in annotator reliability, this would imply a
1805 network with yet more branches and outputs.

The combination of these enhancements with the existing pixel-wise reliability mapping approach could potentially outperform all other techniques, as it would provide the most detailed and accurate representation of annotator reliability while maintaining computational efficiency. The use of data augmentation, in particular, could help the model converge faster and achieve better generalization, as it would expose the model to a wider variety of possible variations in both the input images and the corresponding annotations.

6.3 Academic Discussion

To facilitate the adoption and further development of the proposed approach, several resources have been made publicly available. A comprehensive GitHub repository² has been created that provides easy-to-use APIs for implementing the proposed solution. The repository includes pre-built dataset loaders for TensorFlow/Keras frameworks, implementation of the $TGCE_{SS}$ loss function in all its variants (scalar, features, and pixel-wise), and a neural network architecture builder with experimental examples. The two `tlds` datasets available in the repository are:

- **Triple Negative Breast Cancer (TNBC) Dataset:** A dataset of histopathology images of breast cancer, with annotations of the tumor regions from [?]. The dataset is automatically fetched from AWS S3, which eliminates the need for manual downloads or external dependencies.
- **Oxford-IIIT Pet Dataset:** A dataset of pet images, based on the well-known Oxford-IIIT Pet Dataset with annotations of the pet regions. The mechanism for emulating noisy crowdsourced annotations exposed throughout the present work (see Section ??) is also available in the repository in a tunable manner, just as the TNBC dataset, this `tlds` is batteries included (automatic fetching and mapping, no need to manually download any external source).

²https://github.com/blotero/seg_tgce

To further simplify the adoption of the proposed solution, a PyPI package has been published with detailed documentation available at <https://seg-tgce.readthedocs.io/en/latest/>. The documentation provides comprehensive installation instructions, API reference, usage examples, and best practices for implementation. This package makes it straightforward for the community to integrate the proposed approach into their existing workflows, with a focus on ease of use and clear documentation of all available features and options.

An academic publication is currently being prepared for submission to a Q1 journal. This publication will provide detailed theoretical foundations of the proposed approach, comprehensive experimental results and analysis, comparison with state-of-the-art methods, and discussion of limitations and future directions. The publication will serve as the primary reference for the theoretical and practical aspects of the proposed solution, ensuring that the work is properly documented and validated through the peer-review process.

These resources collectively contribute to the reproducibility and accessibility of the research, enabling the broader scientific community to build upon and extend the proposed approach. The combination of open-source code, well-documented APIs, and peer-reviewed publication ensures that the work can be properly evaluated, validated, and utilized by other researchers in the field. This comprehensive approach to sharing research outputs aligns with current best practices in scientific computing and Machine Learning research, promoting both transparency and practical utility of the proposed solution.

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